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An Optimal Control Problem Arising in Sailboat Trajectory Optimization

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Abstract

We formulate the optimal control problem for a sailboat aiming to reach an upwind buoy, assuming that the wind speed is constant and the wind direction follows a Brownian motion. The yacht navigates in $\mathbb{R}^2$ and the target is a ball of radius $\eta > 0$ centered at the origin. We derive the equations of motion and provide their closed-form solution. The yacht is subject to a tacking cost $c \geq 0$. We define two control problems, one where $c = 0$ and another where $c > 0$. For both cases, we establish an analytical upper bound on the value function by constructing a strategy that enables the yacht to reach the buoy within a finite expected number of tacks. For the case with $c = 0$, where no tacking cost applies, we conjecture an optimal strategy which may involve an infinite number of tacks. The controlled state equation has non-continuous coefficients so standard results on existence and uniqueness of a strong solution do not apply. We provide a proof using the Yamada-Watanabe argument.

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1 Introduction

In this paper we formalize, under a stochastic control perspective, the problem of a sailboat trying to minimize the time to reach an upwind buoy. The motion of the sailboat is influenced by the wind, and on certain time scales, the fluctuations of the wind direction and wind speed can be considered as random. The optimization of the sailboat's trajectory can then be viewed as a stochastic control problem.

In the past years, there has been some scientific literature specific to yacht routing problem. This consists mostly of numerical studies such as the ones in [2, 4]. In these works, the authors use statistical analysis of wind fluctuations, before discretizing the race field, the possible wind directions and the boat's motion in order to solve the the problem of finding the fastest route by stochastic optimization methods. In [5], the authors numerically solve a Markov decision process. There are several works of A. B. Philpott and coauthors, including [10, 12, 13, 11]. In particular [12] presents two models for optimizing yacht routes. The first one is intended for short distance races and the wind is modeled with a Markov chain. The second one treats ocean races and the wind is modeled using long time weather forecasts. Similarly as in [4], they

numerically compute tacking strategies which minimize the expected travel time. More recently, see for example [8] and [17], the problem is formulated with an hybrid control approach and solved numerically with semi-Lagrangian iterative schemes. Another part of works on sailing races that can be found in literature considers the question of winning a race over an opponent, instead of trying to race as fast as possible. This brings into play game theory as in [14, 15]. A simpler, one-dimensional model, is found in [3], where the authors provide a closed form solution.

This is the first work which establishes, to the best of our knowledge, a fully probabilistic approach to the sailboat trajectory optimization problem, without recurring to numerical methods. We propose models of an upwind regatta which capture some of the features of the original problem, but which remains amenable to explicit rigorous solutions that can be proved to be optimal. The results that are presented are extracted from [1, 16].

For a sailboat traveling upwind, one of the most critical choices is deciding when to tack, that is, to change course in such a way that, if the wind was coming from one side of the boat before the tack, it will come from the opposite side afterward. Since this maneuver slows down the yacht, it is natural to model the time lost during a tacking by a "tacking penalty" denoted by c in the sequel. We consider a circular target-buoy that has positive radius $\eta > 0$ and tacking penalty that may be strictly positive or null. In our model, the wind direction fluctuates as a Brownian motion with a fixed volatility σ. The wind speed is assumed constant during the whole race, so is the boat speed, denoted by v.

We begin by doing a coordinates change and aligning the reference frame to the wind direction. This approach enables to define the equations of motion more intuitively and derive a closed-form solution, see Proposition 2.1 and Proposition 2.2. Then, we define the control problems both for the case $c = 0$ and $c > 0$. In the first one, a change of direction of the yacht does not slow it down. The set of *admissible strategies* consists of jointly measurable and adapted processes, see Definition 3.1. Hence, there is a possibility of an infinite number of tacks on any compact set, since they are done at no cost. The second problem introduces a fixed cost for tacking. This changes drastically the sailing strategy and places the mathematical formulation of the problem in the context of optimal stochastic control problems with switching costs. In this last case, *admissible strategies* are piecewise constant processes, see Definition 3.4.

For $c > 0$, we construct a sailing strategy that allows the yacht to reach the target buoy within a time and a number of tacks whose expectation is bounded. We provide an analytical expression for the bound which depends on the speed of the boat and the volatility of the wind, see Theorem 4.14.

In case of no tacking cost, we improve the result of Theorem 4.14 by conjecturing the optimal strategy $A^\star$. Under this strategy, the yacht will reach the target buoy within $\sqrt{2}\frac{r_0-\eta}{v}$ units of time. This bound, therefore, does not depend on the volatility of the wind. The proof that the state equation driven by $A^\star$ admits a unique strong solution does not follow by standard results, because of the lack of continuity of the coefficients, so we provide a new one using the Yamada-Watanabe argument. Optimality of the conjectured optimal strategy will be proved in a companion paper.

The paper is organized as follows. In Section 2 we explain some terminology, perform the change of coordinates and provide the closed form solution to equations of motion. Section 3 formulates the control problems for the two cases $c = 0$ and $c > 0$. In Section 4, we derive the bound on the Value function. Finally, Section 5 deals with the conjectured optimal strategy $A^\star$ for the case $c = 0$.

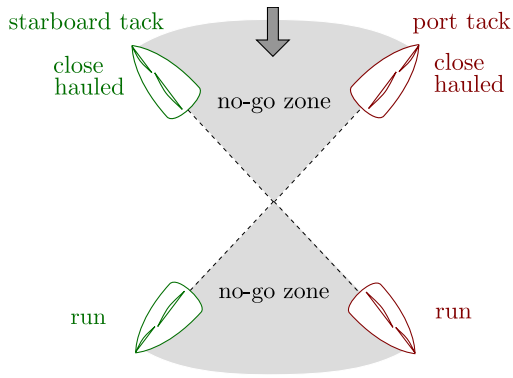

Figure 1: No go zone.

2 Model

In this section, we first explain some basic terminology of sailing that will be used throughout the paper. We then present our model and derive the equations of the position process in both polar and Cartesian coordinates.

2.1 Terminology

If the angle between the boat's heading and the direction of the wind (see Fig. 1) is too narrow, then the boat cannot move forward in this direction. The set of all the angles for which the boat cannot advance is called the *no go zone*. When the wind enters the sails from the right (resp. left) side of the yacht, the yacht is sailing on *starboard tack* (resp. *port tack*). A boat sailing upwind can switch from starboard to port tack, or vice versa, by *tacking*. During a tacking, the yacht enters the no-go zone and hence for a few seconds, it is no longer propelled by the wind. The yacht slows down and it is its inertia that allows it to turn to the other tack. Similarly, if the boat is sailing downwind, then the switch from one tack to the other is called a *jibe*.

For a given wind speed, the velocity of the yacht is a function of the angle between the wind direction and the yacht's heading. This function is illustrated by polar diagrams, see Fig. 2.

When we want to get closer to a particular point, say a buoy, then unless this point lies in the no-go zone with respect to the yacht, we can simply head directly towards it and this determines the tack on which to sail.

However, if the position of the buoy with respect to the yacht is in the no go zone, then the yacht has to follow a zigzag trajectory (see Fig. 3). This is called *beating upwind*: the yacht alternates between routes on starboard and on port tack while trying to sail with the optimal upwind angle, which is the angle in the polar diagram that maximizes the projection of the velocity vector onto the vertical axis (In Fig. 2, this is $\alpha_0 = 45^o$). Likewise, it is not optimal to aim directly at a buoy which lies leeward. The yacht will reach it more quickly by following zigzags of a given angle.

In our model (see Fig. 2), we consider that the yacht's behavior is symmetric if it is beating upwind or running downwind. The optimal angle for an upwind route is fixed at 45°, while for a downwind route, it is 135°. Between these angles, the yacht sails at constant speed $v > 0$ and thus, the polar is a quarter of circle instead of the usual heart shaped curve, see Fig. 2.

When the yacht is beating upwind or running downwind, it must sail at angle α_0 on one

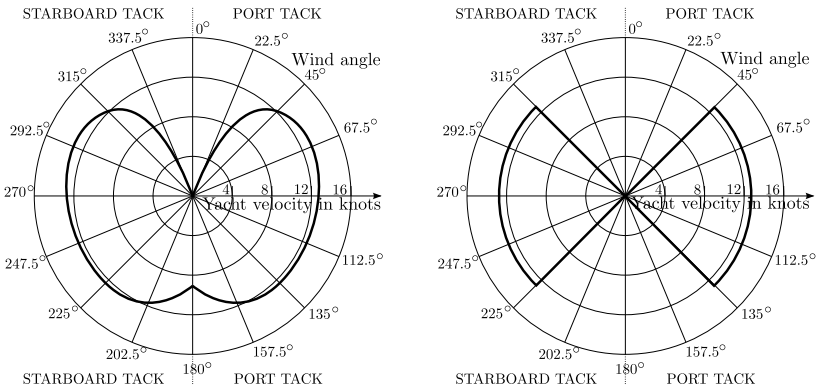

Figure 2: Example of a polar diagram with an 20 knots wind speed (on the left-hand side) and our modeling (on the right-hand side) where the optimal angle for an upwind route is fixed at 45° and for a downwind route, at 135°.

tack, then tack (resp. jibe), continue at angle α_0 on the other tack and repeat this again until the target is reached (see Fig. 3). Many possible tacking strategies can be employed, and in a constant wind, this does not affect the total distance traveled. However, each tack (resp. jibe) impacts the time required. In the present work the exact definition of a strategy and of a best "on average" strategy is going to be given.

In general, both the wind speed and the wind direction change over time. It turns out that moderate fluctuations of wind speed are less relevant than those of wind direction, so we assume that the wind speed is constant. For the wind direction, following [2], we assume that it follows a Brownian motion.

In order to describe the boat's position, for mathematical simplicity, we have chosen to attach the reference frame to the wind instead of to geographic directions. In order to have an understanding of the problem in this new reference frame, assume first that the wind direction is constant and parallel to the vector $(1, 1) \in \mathbb{R}^2$. The motion on starboard (resp. port) tack is shown on the right (resp. left) hand side of Fig. 4, where the yacht is forced to tack (or jibe) on the half line d^1 (resp d^{-1}) shown in the figure (since we will be interested in approaching the origin rather than moving away from it). The grey shaded areas represent the feasible headings of the boat. For each position of the boat and each tack, we choose the direction which points closest to the target. Therefore for each position of the boat, we will only allow two headings, represented by the arrows in Fig. 4, one corresponding to port tack and the other corresponding to starboard tack.

In a fluctuating wind, but with the reference frame attached to the wind, the wind is still represented by a constant vector that is parallel to the vector $(1, 1) \in \mathbb{R}^2$, and it is the boat which is subject both to the displacement induced by sailing and to a Brownian rotation.In this description, a clockwise change of wind direction corresponds to a counterclockwise rotation of the boat's position, and vice-versa.

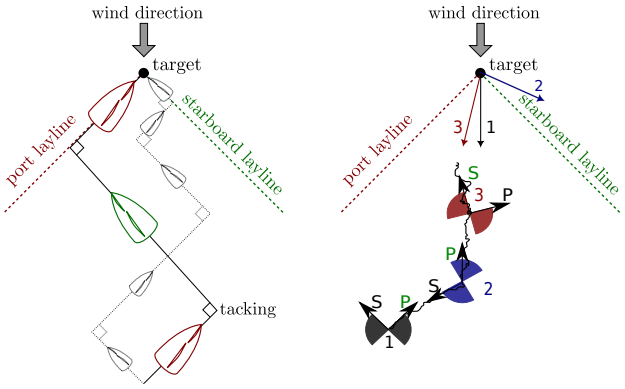

Figure 3: On the left hand side is an example of a tacking strategy for a boat beating upwind under a constant wind. A boat starting at the same point and following any other zigzag parallel to the laylines, would travel the same distance up to the target (dotted trajectory). On the right hand side is an example of a possible trajectory of the boat when the wind direction changes over time as indicated by the arrows at the top of the figure. The right-hand wings (resp. left-hand wings) of the butterfly shaped regions correspond to the feasible directions which can be achieved by sailing on port tack (resp. starboard tack).

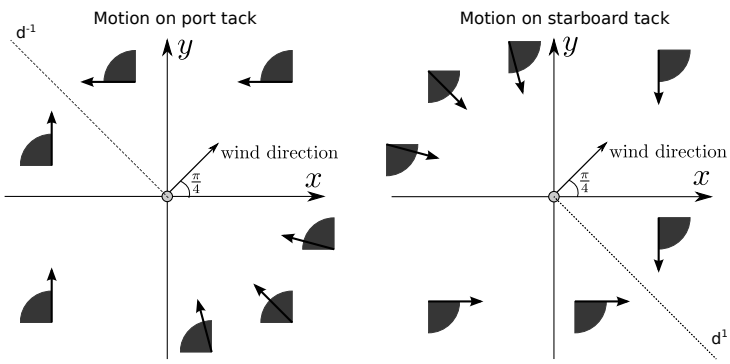

Figure 4: The sailboat's motion on port tack on the left hand side and on starboard tack on the right hand side. Reference frame is attached to the wind direction.

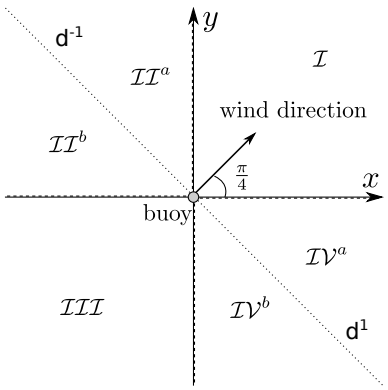

Figure 5: Race area and zones definition.

2.2 Position Process

The purpose of this section is to write the position of the yacht as a 2-dimensional Itô process. Let $\eta > 0$ and define the circular target buoy centered at the origin:

$$\mathbb{B}_\eta := \{(x,y) \in \mathbb{R}^2 : \sqrt{x^2+y^2} \leq \eta\}.$$

In order to describe the motion in our model, we set

$$\begin{aligned} d^1 &= \{(x,y) \in \mathbb{R}^2 : x = -y \text{ and } x > 0\}, \\ d^{-1} &= \{(x,y) \in \mathbb{R}^2 : x = -y \text{ and } x < 0\}, \end{aligned} \tag{2.1}$$

and we split $\mathbb{R}^2_* = \mathbb{R}^2 \setminus \{(0,0)\}$ into several zones (see Fig.5):

$$\begin{aligned} I &= \{(x,y) \in \mathbb{R}^2 : x \geq 0 \text{ and } y \geq 0\} \cap \mathbb{B}^c_\eta, \qquad & II &= (II^a \cup II^b \cup d^{-1}) \cap \mathbb{B}^c_\eta, \\ III &= \{(x,y) \in \mathbb{R}^2 : x \leq 0 \text{ and } y \leq 0\} \cap \mathbb{B}^c_\eta, \qquad & IV &= (IV^a \cup IV^b \cup d^1) \cap \mathbb{B}^c_\eta, \end{aligned}$$

where

$$\begin{aligned} II^a &= \{(x,y) \in \mathbb{R}^2 : x < 0 \text{ and } y > -x\}, \qquad & II^b &= \{(x,y) \in \mathbb{R}^2 : y > 0 \text{ and } y < -x\}, \\ IV^a &= \{(x,y) \in \mathbb{R}^2 : y < 0 \text{ and } y > -x\}, \qquad & IV^b &= \{(x,y) \in \mathbb{R}^2 : x > 0 \text{ and } y < -x\}. \end{aligned}$$

Let $\sigma > 0$ be a given number and $(B_t)_{t\geq 0}$ be a standard real-valued Brownian motion defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$. In our model, B_t represents the wind direction at time t.

Recall (see e.g. [9, p.65]) that if $\vec{\mu}_c$ and $\vec{\Sigma}$ are two functions from $\mathbb{R}^2$ in $\mathbb{R}^2$ defined by

$$\vec{\mu}_c(\vec{x}) = -\tfrac{\sigma^2}{2}\,\vec{x} \qquad \text{and} \qquad \vec{\Sigma}(\vec{x}) = \vec{\Sigma}(x,y) = (-\sigma y, \sigma x),$$

then the unique solution $\vec{X}$ of the system of stochastic differential equations

$$d\vec{X}_t = \vec{\mu}_c(\vec{X}_t)\,dt + \vec{\Sigma}(\vec{X}_t)\,dB_t, \qquad \vec{X}_0 = \vec{x} = (x,y),$$

is a Brownian motion on the circle with speed σ, given by

$$\vec{X}_t = (\cos(\sigma B_t)x - \sin(\sigma B_t)y\,,\, \sin(\sigma B_t)x + \cos(\sigma B_t)y)\,.$$

In the frame attached to the wind, the process $\vec{X}$ defines the position of a motionless yacht.

We now define the functions that correspond to each possible sailing direction of a moving yacht. For $\vec{x} \in \mathbb{R}^2$, set

$$\begin{array}{llll} \vec{\mu}_u(\vec{x}) = (0, v), & \text{(up)} & \vec{\mu}_d(\vec{x}) = (0, -v), & \text{(down)} \\ \vec{\mu}_\ell(\vec{x}) = (-v, 0), & \text{(left)} & \vec{\mu}_r(\vec{x}) = (v, 0), & \text{(right)} \\ \vec{\mu}_\rho(\vec{x}) = -v\frac{\vec{x}}{\|\vec{x}\|}. & \text{(radial)} & & \end{array}$$

The function corresponding to the starboard tack (resp. port tack) is denoted as $\vec{\mu}(\vec{x}, 1)$ (resp $\vec{\mu}(\vec{x}, -1)$). According to Fig. 4 and Fig. 5, we have

$$\begin{aligned} \vec{\mu}(\vec{x}, 1) &= \vec{\mu}_d(\vec{x})\mathbf{1}_{\{I \cup IV^a\}}(\vec{x}) + \vec{\mu}_r(\vec{x})\mathbf{1}_{\{III \cup IV^b\}}(\vec{x}) + \vec{\mu}_\rho(\vec{x})\mathbf{1}_{\{II\}}(\vec{x}) + \vec{\mu}_c(\vec{x}), \\ \vec{\mu}(\vec{x}, -1) &= \vec{\mu}_\ell(\vec{x})\mathbf{1}_{\{I \cup II^a\}}(\vec{x}) + \vec{\mu}_u(\vec{x})\mathbf{1}_{\{III \cup II^b\}}(\vec{x}) + \vec{\mu}_\rho(\vec{x})\mathbf{1}_{\{IV\}}(\vec{x}) + \vec{\mu}_c(\vec{x}). \end{aligned} \tag{2.2}$$

The position process of a yacht sailing on starboard tack, respectively on port tack, is now the solution of the following system of SDEs:

$$d\vec{X}_t = \vec{\mu}(\vec{X}_t, 1)\, dt + \vec{\Sigma}(\vec{X}_t)\, dB_t, \qquad \vec{X}_0 = \vec{x}_0, \tag{2.3}$$

respectively,

$$d\vec{X}_t = \vec{\mu}(\vec{X}_t, -1)\, dt + \vec{\Sigma}(\vec{X}_t)\, dB_t, \qquad \vec{X}_0 = \vec{x}_0. \tag{2.4}$$

The next two propositions lead to the solution of these SDEs.

Proposition 2.1. *Let $\vec{x} = (x, y) \in \mathbb{R}^2$ and $v_1, v_2 \in \mathbb{R}$. The process $(\vec{X}_t = (X_t, Y_t))$, $t \in [0, +\infty[$, taking values in $\mathbb{R}^2$ and defined by*

$$\vec{X}_t = \begin{pmatrix} \cos(\sigma B_t)x - \sin(\sigma B_t)y - v_1 \displaystyle\int_0^t \cos(\sigma B_t - \sigma B_s)ds + v_2 \int_0^t \sin(\sigma B_t - \sigma B_s)ds \\ \sin(\sigma B_t)x + \cos(\sigma B_t)y - v_1 \displaystyle\int_0^t \sin(\sigma B_t - \sigma B_s)ds - v_2 \int_0^t \cos(\sigma B_t - \sigma B_s)ds \end{pmatrix} \tag{2.5}$$

is the unique solution of

$$\begin{cases} dX_t &= \left(-\frac{\sigma^2}{2} X_t - v_1\right) dt - \sigma Y_t\, dB_t, \qquad X_0 = x, \\ dY_t &= \left(-\frac{\sigma^2}{2} Y_t - v_2\right) dt + \sigma X_t\, dB_t, \qquad Y_0 = y. \end{cases} \tag{2.6}$$

Proof. Uniqueness follows directly from [9, Thm. 5.2.1], since the drift, as well as the diffusion coefficient in (2.6), are affine. Thus, it only remains to verify that the process $\vec{X}$ defined by (2.5) is indeed a solution of (2.6) using Itô's formula. For details see [16, Prop. 5.1]. □

Proposition 2.2. *Let $\vec{x} = (x, y) \in \mathbb{R}^2_*$. The process $\left(\vec{X}_t,\, t \in \left[0, \frac{\|\vec{x}\|}{v} - \eta\right]\right)$ taking values in $\mathbb{R}^2 \cap \mathbb{B}^c_\eta$ defined by*

$$\vec{X}_t = \begin{pmatrix} X_t \\ Y_t \end{pmatrix} = \begin{pmatrix} (\|\vec{x}\| - vt)\cos(\sigma B_t + \theta) \\ (\|\vec{x}\| - vt)\sin(\sigma B_t + \theta) \end{pmatrix} \tag{2.7}$$

where $\theta \in \mathbb{R}$ is such that $\cos(\theta) = \frac{x}{\|\vec{x}\|}$ and $\sin(\theta) = \frac{y}{\|\vec{x}\|}$, is the unique solution of

$$\begin{cases} dX_t &= \left(-\frac{\sigma^2}{2} X_t - \frac{vX_t}{\|\vec{X}_t\|}\right) dt - \sigma Y_t\, dB_t, \qquad X_0 = x, \\ dY_t &= \left(-\frac{\sigma^2}{2} Y_t - \frac{vY_t}{\|\vec{X}_t\|}\right) dt + \sigma X_t\, dB_t, \qquad Y_0 = y. \end{cases} \tag{2.8}$$

Proof. Observe that the drift coefficient in (2.8) is Lipschitz continuous in $\mathbb{R}^2 \cap \mathbb{B}_\eta^c$, hence the existence and uniqueness of the solution holds on $t \in \left[0, \frac{\|\vec{x}\|}{v} - \eta\right]$. It remains to verify that the process $\vec{X}$ defined by (2.7) is indeed a solution of (2.8). For details see [16, Prop. 5.2]. □

We see that the system (2.8) can also be written

$$d\vec{X}_t = (\vec{\mu}_c(\vec{X}_t) + \vec{\mu}_\rho(\vec{X}_t))\, dt + \sigma(\vec{X}_t)\, dB_t, \qquad \vec{X}_0 = \vec{x}, \quad t \in \left[0, \frac{\|\vec{x}\|}{v}\right],$$

which corresponds to the equation satisfied by the position process of a yacht sailing radially in direction of the buoy. This happens if the boat is sailing on port tack in zone *IV* or on starboard tack in zone *II*.

On the other hand, Proposition 2.1 gives us the system of SDEs and the corresponding solutions for the other four motions:

(1) downward motion ($v_1 = 0, v_2 = v$) when the boat is on starboard tack in zone $I \cup IV^a$,

(2) upward motion ($v_1 = 0, v_2 = -v$) when the boat is on port tack in zone $II^b \cup III$,

(3) leftward motion ($v_1 = v, v_2 = 0$) when the boat is on port tack in zone $I \cup II^a$,

(4) rightward motion ($v_1 = -v, v_2 = 0$) when the boat is on starboard tack in zone $III \cup IV^b$.

Therefore, Prop. 2.1 and Prop. 2.2 provide explicit formulas to describe the solution of (2.3) and (2.4).

Remark 2.3. Denote (X_t^1, Y_t^1) (resp. (X_t^{-1}, Y_t^{-1})) the process defined by (2.3) (resp. (2.4)). Define $\tau_f^a := \inf\{t : (X_t^a, Y_t^a) \in d^a \cup \mathbb{B}_\eta\}$, $a \in \{-1, +1\}$ where d^1, and d^{-1} are defined in (2.1). Equation (2.3) (resp. (2.4)) have Lipschitz continuous coefficients in $\mathbb{R}^2 \setminus d^1$ (resp. $\mathbb{R}^2 \setminus d^{-1}$), therefore by a standard localization argument, there exists a strong solution until time τ_f^1 (resp. τ_f^{-1}), for any initial condition $\bar{x}_0 = (x_0, y_0) \in \mathbb{R}^2 \setminus d^1$ (resp $\bar{x}_0 = (x_0, y_0) \in \mathbb{R}^2 \setminus d^{-1}$).

For later use, the equations of motion are also written in radial coordinates. For definiteness, we consider angles in $\left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[$ and the general case is then obtained by periodic extension. Let us consider, for instance, the process describing a yacht sailing downwind. Setting $v_1 = 0$ and $v_2 = v$ in (2.5), we see that the position process $\vec{X}$ satisfies

$$\begin{cases} dX_t &= -\frac{\sigma^2}{2} X_t\, dt - \sigma Y_t\, dB_t, & X_0 = x, \\ dY_t &= \left(-\frac{\sigma^2}{2} Y_t - v\right) dt + \sigma X_t\, dB_t, & Y_0 = y. \end{cases}$$

Applying Itô's formula to $R_t = \sqrt{(X_t)^2 + (Y_t)^2}$ and $\Theta_t = \arctan(Y_t/X_t)$, we get, after simplification,

$$R_t = R_0 - \int_0^t v \frac{Y_s}{R_s}\, ds = R_0 - \int_0^t v \sin(\Theta_s)\, ds$$

and

$$\Theta_t = \Theta_0 - \int_0^t \frac{v \cos(\Theta_s)}{R_s}\, ds + \int_0^t \sigma\, dB_s.$$

In other words, the process (R_t, Θ_t) satisfies

$$\begin{cases} dR_t &= -v \sin(\Theta_t)\, dt, & R_0 = \|\vec{x}\|, \\ d\Theta_t &= -\dfrac{v \cos(\Theta_t)}{R_t}\, dt + \sigma\, dB_t, & \Theta_0 = \arctan(y/x). \end{cases}$$

We do the same for all other sailing directions and we find that *the position of the yacht in polar coordinates* $(R_t^a, \Theta_t^a)_{t\geq 0}$, for $a \in \{-1, 1\}$, is the solution of

$$\begin{cases} dR_t^a = \mu_1(\Theta_t^a, a)\, dt, \\ d\Theta_t^a = \mu_2(R_t^a, \Theta_t^a, a)\, dt + \sigma\, dB_t, \end{cases} \tag{2.9}$$

on $\{t < \tau_f^a\}$, where τ_f^a is defined in Remark 2.3, and $\vec{\mu}(r, \theta, a) = (\mu_1(\theta, a), \mu_2(r, \theta, a))$ is given, for $\theta \in [-\frac{\pi}{4}, \frac{7\pi}{4}[$, by

$$\vec{\mu}(r, \theta, 1) = \vec{\mu}_d(r, \theta)\mathbf{1}_{\{\theta\in[-\frac{\pi}{4}, \frac{\pi}{2}[\}} + \vec{\mu}_\rho(r, \theta)\mathbf{1}_{\{\theta\in[\frac{\pi}{2}, \pi[\}} + \vec{\mu}_r(r, \theta)\mathbf{1}_{\left\{\theta\in\left[\pi, \frac{7\pi}{4}\right[\right\}}, \tag{2.10}$$

$$\vec{\mu}(r, \theta, -1) = \vec{\mu}_\ell(r, \theta)\mathbf{1}_{\left\{\theta\in\left[0, \frac{3\pi}{4}\right[\right\}} + \vec{\mu}_u(r, \theta)\mathbf{1}_{\left\{\theta\in\left[\frac{3\pi}{4}, \frac{3\pi}{2}\right[\right\}} + \vec{\mu}_\rho(r, \theta)\mathbf{1}_{\left\{\theta\in[-\frac{\pi}{4}, 0[\cup\left[\frac{3\pi}{2}, \frac{7\pi}{4}\right[\right\}} \tag{2.11}$$

where

$$\begin{aligned} \vec{\mu}_d(r, \theta) &= \left(-v\sin\theta, -v\tfrac{\cos\theta}{r}\right), & \vec{\mu}_\ell(r, \theta) &= \left(-v\cos\theta, v\tfrac{\sin\theta}{r}\right), \\ \vec{\mu}_u(r, \theta)) &= \left(v\sin\theta, v\tfrac{\cos\theta}{r}\right), & \vec{\mu}_r(r, \theta) &= \left(v\cos\theta, -v\tfrac{\sin\theta}{r}\right), \\ \vec{\mu}_\rho(r, \theta) &= (-v, 0), \end{aligned}$$

and extended to $\theta \in \mathbb{R}$ by 2π-periodicity. The functions μ_1 and μ_2 appearing in (2.9) can be written more explicitly, using (2.10)–(2.11). For the motion on *starboard tack* and $\theta \in [-\frac{\pi}{4}, \frac{7\pi}{4}[$, they are defined as:

$$\mu_1(\theta, 1) = \begin{cases} -v\sin(\theta), \\ -v, \\ v\cos(\theta), \end{cases} \qquad \mu_2(r, \theta, 1) = \begin{cases} -\frac{v\cos(\theta)}{r}, & \theta \in [-\frac{\pi}{4}, \frac{\pi}{2}[, \\ 0, & \theta \in [\frac{\pi}{2}, \pi[, \\ -\frac{v\sin(\theta)}{r}, & \theta \in [\pi, \frac{7\pi}{4}[. \end{cases} \tag{2.12}$$

Similarly, for the motion of the boat on *port tack* and $\theta \in [-\frac{\pi}{4}, \frac{7\pi}{4}[$, they are equal to:

$$\mu_1(\theta, -1) = \begin{cases} -v, \\ -v\cos(\theta), \\ v\sin(\theta), \end{cases} \qquad \mu_2(r, \theta, -1) = \begin{cases} 0, & \theta \in [\frac{3\pi}{2}, \frac{7\pi}{4}[\cup[-\frac{\pi}{4}, 0[, \\ \frac{v\sin(\theta)}{r}, & \theta \in [0, \frac{3\pi}{4}[, \\ \frac{v\cos(\theta)}{r}, & \theta \in [\frac{3\pi}{4}, \frac{3\pi}{2}[. \end{cases}$$

Therefore, after extending to $\theta \in \mathbb{R}$ by 2π-periodicity the coefficients μ_1 and μ_2 above, the functions below are defined:

$$\begin{aligned} &\mu_1(\theta, a) : \mathbb{R} \times \{-1, +1\} \mapsto [-v, v], \\ &\mu_2(r, \theta, a) : \mathbb{R}_+ \times \mathbb{R} \times \{-1, +1\} \mapsto \mathbb{R}. \end{aligned} \tag{2.13}$$

See Fig. 6 for the illustration of the sailing rules in polar coordinates.

Remark 2.4. A strong solution to (2.9) up to time τ_f^a is well defined because of the corresponding result in Cartesian coordinates (see Remark 2.3).

3 Control Problem

In this section we introduce the control problem and prove a Verification theorem for the case $c = 0$. In view of (2.9), we define the state equation as:

$$\begin{cases} dR_t^A = 1_{\{R_t^A>0\}}\mu_1(\Theta_t^A, A_t)dt, \\ d\Theta_t^A = 1_{\{R_t^A>0\}}\mu_2(R_t^A, \Theta_t^A, A_t)dt + \sigma 1_{\{R_t^A>0\}}dB_t, \end{cases} \tag{3.1}$$

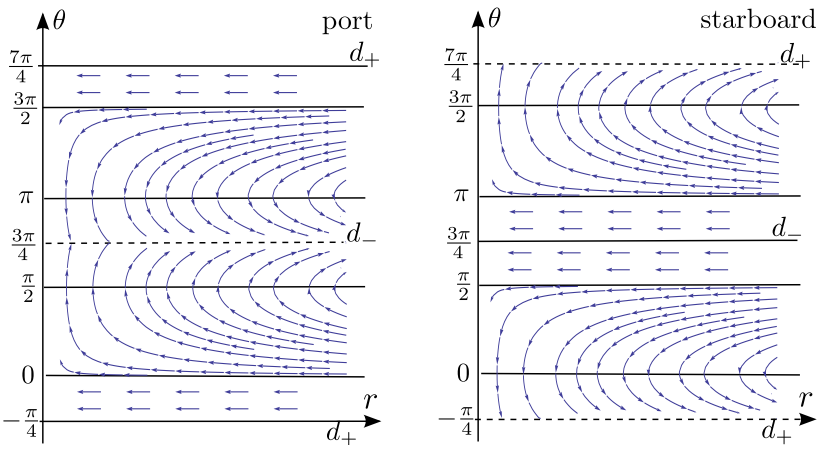

Figure 6: Sailing rules in polar coordinates. On the left-hand side, the yacht is on port tack ($a = -1$). On the right-hand side, the yacht is on starboard tack ($a = 1$). The flow of arrows indicates the direction of the drift. In addition to the motion along the flow, the yacht moves as a Brownian motion in a direction parallel to the θ-axis, with a speed that depends on the diffusion coefficient σ. Notice that there is no Brownian component in the radial direction.

and the time to reach the target buoy $\mathbb{B}_\eta$:

$$\tau_\eta^A := \inf\left\{t : R_t^A \leq \eta\right\}. \tag{3.2}$$

The process $(A_t)_{t\in[0,+\infty[}$ is an *admissible strategy*. Depending on the case $c = 0$ or $c > 0$ the definition of admissible strategy is different and, respectively, stated in Definition 3.1 or Definition 3.4.

The position of the yacht at time t driven by A is defined as the $\mathbb{P}_{r,\theta,a}$ strong solution $(R_t^A, \Theta_t^A)_{t\in[0,\tau_\eta^A]}$ of (3.1). where $\mathbb{P}_{r,\theta,a}$ denotes a family of probability measures such that

$$\mathbb{P}_{r,\theta,a}\left\{(R_0^A, \Theta_0^A) = (r,\theta),\ A_{0-} = a\right\} = 1. \tag{3.3}$$

The initial tack a is the yacht's tack just before the race starts, so $A_0 \neq A_{0-}$ if the yacht tacks immediately after the race starts.

3.1 Case $c = 0$

A strategy which allows an infinite number of tacks might reach $\mathbb{B}_\eta$ in a finite time. Since $c = 0$, the initial tack of the boat has no influence at all on the problem, hence we will omit it in (3.3), and write $\mathbb{P}_{r,\theta}$.

Definition 3.1. Let $\eta > 0$, $(r,\theta) \in \mathbb{R}_+ \times \mathbb{R}$ and $A : (\mathbb{R}_+ \cup \{0-\}) \times \Omega \mapsto \{+1, -1\}$ be a jointly measurable and adapted process in (t,ω). We say that A is an *admissible strategy* if:

1. equation (3.1) admits a strong solution for $t \in [0, \tau_\eta^A]$,
2. $\mathbb{E}_{r,\theta}\left(\tau_\eta^A\right) < \infty$.

The set of admissible strategies under $\mathbb{P}_{r,\theta}$ is denoted by $\mathcal{A}_{r,\theta}^\eta$ (often the subscript $\cdot_{r,\theta}$ will be omitted).

In Section 4 we shall prove that, for every $\eta > 0$, the set $\mathcal{A}^\eta$ is non empty.

The control problem $\mathcal{P}(\eta)$ consists in finding an *admissible strategy* that minimizes the expected time to reach the upwind buoy. Let, for $A \in \mathcal{A}^\eta$,

$$J^\eta(r, \theta, A) = \mathbb{E}_{r,\theta}\left(\tau_\eta^A\right),$$

Then the *Value function* is defined as

$$V^\eta(r, \theta) = \inf_{A \in \mathcal{A}^\eta} J^\eta(r, \theta, A). \tag{3.4}$$

In the above definition, $V^\eta : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$ and $J^\eta : \mathbb{R}_+ \times \mathbb{R} \times \mathcal{A}^\eta \to \mathbb{R}_+$, hence the control problem consists in finding an *admissible strategy* $A^\star$ which is optimal, that is $V^\eta(r, \theta) = J^\eta(r, \theta, A^\star)$.

The theorem below is the so called Verification Theorem. Having a candidate Value function $\tilde{V}^\eta$, and a strategy $A^\star \in \mathcal{A}^\eta$ that is guessed to be optimal, it provides a sufficient condition for $\tilde{V}^\eta$ to be the Value function of the problem $\mathcal{P}(\eta)$ and for the strategy $A^\star$ to be optimal. Let N^η be defined in the following way:

$$N^\eta : \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+, \qquad N^\eta(t, r, \theta) := t + \tilde{V}^\eta(r, \theta),$$

for some function $\tilde{V}^\eta(r, \theta) : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$.

Theorem 3.2. *Fix $\eta > 0$ and suppose there exists a continuous function $\tilde{V}^\eta(r, \theta) : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$ such that*

(a) $\tilde{V}^\eta(\eta, \theta) = 0$ for all $\theta \in \mathbb{R}$,

(b) $\tilde{V}^\eta$ is continuous at $r = \eta$, uniformly in $\theta \in \mathbb{R}$,

(c) there exist $C, D \in \mathbb{R}_+$, such that for all $r \geq 0$, we have $\tilde{V}^\eta(r, \theta) \leq Cr + D$.

Let (R_t^A, Θ_t^A) be the process defined in (3.1). If

(1) for all admissible strategies $A \in \mathcal{A}^\eta$,

$$N^\eta(t \wedge \tau_\eta^A, R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A}) = t \wedge \tau_\eta^A + \tilde{V}^\eta(R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A})$$

is a submartingale, then $\tilde{V}^\eta(r, \theta) \leq V^\eta(r, \theta)$;

(2) if, in addition, there is a strategy $A^\star \in \mathcal{A}^\eta$ for which

$$N^\eta(t \wedge \tau_\eta^A, R^{A^\star}_{t\wedge\tau_\eta^{A^\star}}, \Theta^{A^\star}_{t\wedge\tau_\eta^{A^\star}}) = t \wedge \tau_\eta^{A^\star} + \tilde{V}^\eta(R^{A^\star}_{t\wedge\tau_\eta^{A^\star}}, \Theta^{A^\star}_{t\wedge\tau_\eta^{A^\star}})$$

is a martingale, then $\tilde{V}^\eta(r, \theta) = V^\eta(r, \theta)$.

Proof. Let r and θ be fixed.

(1) We recall that an admissible strategy $A \in \mathcal{A}^\eta$ has the property that $\mathbb{E}\left(\tau_\eta^A\right) < +\infty$.

The submartingale property of $N^\eta(t \wedge \tau_\eta^A, R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A})$ implies, since $t \wedge \tau_\eta^A$ is a bounded stopping time,

$$\begin{aligned}\tilde{V}^\eta(r, \theta) &= \mathbb{E}\left(N^\eta(0, r, \theta)\right) \leq \mathbb{E}\left(N^\eta(t \wedge \tau_\eta^A, R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A})\right) \\ &= \mathbb{E}\left((t \wedge \tau_\eta^A) + \tilde{V}^\eta(R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A})\right).\end{aligned} \tag{3.5}$$

Since, by definition of admissible strategy, $\tau_\eta^A < \infty$ $\mathbb{P}$-a.s., then $t \wedge \tau_\eta^A \to \tau_\eta^A$ a.s. as $t \to +\infty$. By the monotone convergence theorem, as $t \to \infty$,

$$\mathbb{E}\left(t \wedge \tau_\eta^A\right) \to \mathbb{E}\left(\tau_\eta^A\right). \tag{3.6}$$

Observe that $R^A_{t\wedge\tau_\eta^A} \to R^A_{\tau_\eta^A} = \eta$ as $t \to +\infty$. By the assumption of uniform continuity at $r = \eta$, and since $V^\eta(\eta, \cdot) = 0$, then $V^\eta(R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A}) \to 0$ as $t \to +\infty$.

By (2.13), we have that $|\mu_1(\theta)| \leq v$ for $\theta \in \mathbb{R}$. It follow that

$$|R^A_{t\wedge\tau_\eta^A} - r| \leq v(t \wedge \tau_\eta^A) \leq v\tau_\eta^A,$$

so

$$R^A_{t\wedge\tau_\eta^A} \leq r + v\tau_\eta^A.$$

By assumption (c):

$$\tilde{V}^\eta(R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A}) \leq CR_{t\wedge\tau_\eta^A} + D \leq Cv\tau_\eta^A + Cr + D.$$

Since $\mathbb{E}\left(\tau_\eta^A\right) < \infty$, we can apply the dominated convergence theorem to see that, as $t \to +\infty$,

$$\mathbb{E}\left(\tilde{V}^\eta(R^A_{t\wedge\tau_\eta^A}, \Theta^A_{t\wedge\tau_\eta^A})\right) \to 0. \tag{3.7}$$

From (3.5), (3.6), (3.7), we conclude that

$$\tilde{V}^\eta(r, \theta) \leq \mathbb{E}\left(\tau_\eta^A\right) = J^\eta(r, \theta, A).$$

Since this holds for all $A \in \mathcal{A}^\eta$, by taking the infimum over A it follows

$$\tilde{V}^\eta(r, \theta) \leq V^\eta(r, \theta). \tag{3.8}$$

(2) Under the strategy $A^\star$, (3.5) holds with the inequality replaced by an equality, hence

$$\tilde{V}^\eta(r, \theta) = \mathbb{E}\left(t \wedge \tau_\eta^{A^\star} + \tilde{V}^\eta(R^{A^\star}_{t\wedge\tau_\eta^{A^\star}}, \Theta^{A^\star}_{t\wedge\tau_\eta^{A^\star}})\right).$$

Using the same argument as in Item (1),

$$\lim_{t\to\infty} \mathbb{E}\left(t \wedge \tau_\eta^{A^\star} + \tilde{V}^\eta(R^{A^\star}_{t\wedge\tau_\eta^{A^\star}}, \Theta^{A^\star}_{t\wedge\tau_\eta^{A^\star}})\right) = \mathbb{E}\left(\tau_\eta^{A^\star}\right).$$

This implies

$$\tilde{V}^\eta(r, \theta) = \mathbb{E}\left(\tau_\eta^{A^\star}\right) \geq \inf_{A\in\mathcal{A}^\eta} \mathbb{E}\left(\tau_\eta^A\right) = V^\eta(r, \theta),$$

and by (3.8), the conclusion follows.

□

3.2 Case $c > 0$

This is the case where the tacking cost is positive. Here a strategy cannot allow an infinite number of tacks, hence it should be piecewise constant.

Definition 3.3. Let $X_t : [0, \infty[\times\Omega \to \mathbb{R}^n$ be an $(\mathcal{F}_t)_{t\geq 0}$ adapted and right continuous process. If, for every $\omega \in \Omega$ and $T < +\infty$, there exists ε such that for all $t \in [0, T] : X_t \neq X_{t-}$, then $X_{r+t} = X_t$ for $r \in [0, \varepsilon]$, we say that X is a *piecewise constant process.* The *jump* of X_t at $t > 0$ is defined as $\Delta X_t = X_t - X_{t-}$.

The yacht spends c units of time when tacking. The total amount of time spent in tacking needs to be integrable for a strategy to be admissible. The main difference between Definition 3.4 and the definition of admissible strategy (see Definition 3.1) given in the case $c = 0$ is the restriction to piecewise constant processes and the condition on integrability of the number of tacks, formulated in Item (c) below.

Definition 3.4. Let $\eta > 0$, $c > 0$, $(r, \theta) \in \mathbb{R}_+ \times \mathbb{R}$ and $a \in \{-1, +1\}$. Let $A : (\mathbb{R}_+ \cup \{0-\}) \times \Omega \mapsto \{+1, -1\}$ be a piecewise constant process such that:

(a) there exists a $\mathbb{P}_{r,\theta,a}$ strong solution to (3.1) in $[0, \tau_\eta^A]$, where τ_η^A is defined in (3.2),

(b) $\mathbb{E}_{r,\theta,a}\left(\tau_\eta^A\right)$ is finite,

(c) $\mathbb{E}_{r,\theta,a}\left(M_{\tau_\eta^A}(A)\right) < \infty$, where

$$M_t(A) = \sum_{s=0}^{t} \Delta A_s.$$

The process A is called an *admissible strategy.* The set of the admissible strategies under $\mathbb{P}_{r,\theta,a}$ (defined in (3.3)) is denoted by $\mathcal{A}_{r,\theta,a}^{c,\eta}$ (often the subscript $\cdot_{r,\theta,a}$ will be omitted).

In Section 4 we show that $\mathcal{A}^{c,\eta}$ is non empty.

Remark 3.5. Observe that, for $\eta > 0$, $\mathcal{A}^{c,\eta} \subset \mathcal{A}^\eta$ where $\mathcal{A}^\eta$ is introduced in Definition 3.1.

The control problem $\mathcal{P}(c, \eta)$ consists in finding an *admissible strategy* that minimizes the time to reach the upwind buoy. Letting

$$J^{c,\eta}(r, \theta, a, A) = \mathbb{E}_{r,\theta,a}\left(\tau_\eta^A + cM_{\tau_\eta^A}(A)\right) \tag{3.9}$$

then the *Value function* is defined as

$$V^{c,\eta}(r, \theta, a) = \inf_{A\in\mathcal{A}^{c,\eta}} J^\eta(r, \theta, a, A). \tag{3.10}$$

In the above definition, $V^{c,\eta} : \mathbb{R}_+ \times \mathbb{R} \times \{-1, +1\} \to \mathbb{R}_+$ and $J^{c,\eta} : \mathbb{R}_+ \times \mathbb{R} \times \{-1, +1\} \times \mathcal{A}^{c,\eta} \to \mathbb{R}_+$, hence the control problem $\mathcal{P}(c, \eta)$ consists in finding an *admissible strategy* $A^\star \in \mathcal{A}^{c,\eta}$ such that $V^{c,\eta}(r, \theta, a) = J^{c,\eta}(r, \theta, a, A^\star)$.

4 Upper bound on the Value function

In this section, we derive an analytical expression for an upper bound on the Value function (3.10). The difficulty is to prove that it is possible to reach the buoy with a finite number of tackings. Firstly, we define a particular tacking strategy and prove that the payoff (3.9) for this

strategy is bounded for any starting point. Theorem 4.14, which is the main result of the section, will be then a direct consequence of this.

Let $\alpha \in \left]0, \frac{\pi}{8}\right]$ and $r_0 > \eta$. We define

$$\begin{aligned} D_1^1 &= \left([\eta, r_0[\times \left(\left[-\tfrac{\pi}{4}, -\alpha\right] \cup \left[\tfrac{3\pi}{2} + \alpha, \tfrac{7\pi}{4}\right[\right)\right) &&, \quad D_2^1 = \left(\{r_0\} \times \left(\left[-\tfrac{\pi}{4}, 0\right] \cup \left[\tfrac{3\pi}{2}, \tfrac{7\pi}{4}\right[\right)\right); \\ D_1^{-1} &= \left([\eta, r_0[\times \left[\tfrac{\pi}{2} + \alpha, \pi - \alpha\right]\right) &&, \quad D_2^{-1} = \left(\{r_0\} \times \left[\tfrac{\pi}{2}, \pi\right]\right) \end{aligned}$$

and

$$D^1 = D_1^1 \cup D_2^1 \qquad , \quad D^{-1} = D_1^{-1} \cup D_2^{-1}.$$

Let C^1 (resp. C^{-1}) be the complement sets of D^1 (resp. D^{-1}) in $[\eta, r_0] \times \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[$ (see Fig. 7). We denote by $\bar{C}^a$ the closure of C^a. The idea of the strategy A defined below is that as soon as the yacht touches the set D^1 (resp. D^{-1}) and is on starboard (resp. port) tack then it should tack. This is consistent with the definition of the forbidden lines d^1 and d^{-1}, see (2.1).

Suppose the initial position of the boat is (r, θ), $\theta \in]-\frac{\pi}{4}, \frac{7\pi}{4}[$, with initial tack $a \in \{-1, +1\}$. Let the strategy A be defined as $A_{0-} = a$, and

$$\begin{cases} A_0 = -1 & \text{if } (r, \theta) \in D^1, \\ A_0 = 1 & \text{if } (r, \theta) \in D^{-1}, \\ A_0 = a & \text{elsewhere.} \end{cases}$$

Define $(R_t^{(1)}, \Theta_t^{(1)})$ by $(R_0^{(1)}, \Theta_0^{(1)}) = (r, \theta)$, and for $t \in [0, \psi^{(1)}[$,

$$\begin{cases} dR_t^{(1)} &= \mu_1(\Theta_t^{(1)}, A_0)dt, \\ d\Theta_t^{(1)} &= \mu_2(R_t^{(1)}, \Theta_t^{(1)}, A_0)dt + \sigma dB_t, \end{cases}$$

where

$$\begin{aligned} \psi^{(1)} &:= \nu^{(1)} \wedge \rho^{(1)}, \\ \nu^{(1)} &:= \inf\left\{t \geq 0 : \ (R_t^{(1)}, \Theta_t^{(1)}) \in D_1^{A_0}\right\} \wedge \tau_\eta^{(1)}, \\ \rho^{(1)} &:= \inf\left\{t \geq 0 : \ (R_t^{(1)}, \Theta_t^{(1)}) \in D_2^{A_0}\right\} \wedge \tau_\eta^{(1)}, \\ \tau_\eta^{(1)} &:= \inf\left\{t \geq 0 : \ R_t^{(1)} = \eta\right\} \text{ if } \{\cdots\} \neq \emptyset \text{ else } +\infty. \end{aligned}$$

We recall that the coefficients μ_1 and μ_2 are defined in (2.13). Then define $(R_t^{(2)}, \Theta_t^{(2)})$ by $(R_{\psi^{(1)}}^{(2)}, \Theta_{\psi^{(1)}}^{(2)}) = (R_{\psi^{(1)}}^{(1)}, \Theta_{\psi^{(1)}}^{(1)})$, and for $t \in [\psi^{(1)}, \psi^{(2)}[$,

$$\begin{cases} dR_t^{(2)} &= \mu_1(\Theta_t^{(2)}, -A_0)dt, \\ d\Theta_t^{(2)} &= \mu_2(R_t^{(2)}, \Theta_t^{(2)}, -A_0)dt + \sigma dB_t, \end{cases}$$

where

$$\begin{aligned} \psi^{(2)} &:= \nu^{(2)} \wedge \rho^{(2)}, \\ \nu^{(2)} &:= \inf\left\{t \geq \psi^{(1)} : \ (R_t^{(2)}, \Theta_t^{(2)}) \in D_1^{-A_0}\right\} \wedge \tau_\eta^{(2)}, \\ \rho^{(2)} &:= \inf\left\{t \geq \psi^{(1)} : \ (R_t^{(2)}, \Theta_t^{(2)}) \in D_2^{-A_0}\right\} \wedge \tau_\eta^{(2)}, \\ \tau_\eta^{(2)} &:= \inf\left\{t \geq \psi^{(1)} : \ R_t^{(2)} = \eta\right\} \text{ if } \{\cdots\} \neq \emptyset \text{ else } +\infty. \end{aligned}$$

We remark that, if $\psi^{(1)} = \tau_0^{(1)}$, then also $\psi^{(2)} = \tau_\eta^{(2)} = \tau_\eta^{(1)}$.

Assuming $(R_t^{(i-1)}, \Theta_t^{(i-1)})$ and $\psi^{(i-1)}, \nu^{(i-1)}, \rho^{(i-1)}, \tau_0^{(i-1)}$ have been defined, let $(R^{(i)}_{\psi^{(i-1)}}, \Theta^{(i)}_{\psi^{(i-1)}}) = (R^{(i-1)}_{\psi^{(i-1)}}, \Theta^{(i-1)}_{\psi^{(i-1)}})$, and for $t \in [\psi^{(i-1)}, \psi^{(i)}[$:

$$\begin{cases} dR_t^{(i)} &= \mu_1(\Theta_t^{(i)}, (-1)^{i-1}A_0)dt, \\ d\Theta_t^{(i)} &= \mu_2(R_t^{(i)}, \Theta_t^{(i)}, (-1)^{i-1}A_0)dt + \sigma dB_t, \end{cases}$$

and

$$\begin{aligned} \psi^{(i)} &:= \nu^{(i)} \wedge \rho^{(i)}, \\ \nu^{(i)} &:= \inf\left\{t \geq \psi^{(i-1)} : (R_t^{(i)}, \Theta_t^{(i)}) \in D_1^{(-1)^{i-1}A_0}\right\} \wedge \tau_\eta^{(i)}, \\ \rho^{(i)} &:= \inf\left\{t \geq \psi^{(i-1)} : (R_t^{(i)}, \Theta_t^{(i)}) \in D_2^{(-1)^{i-1}A_0}\right\} \wedge \tau_\eta^{(i)}, \\ \tau_\eta^{(i)} &:= \inf\left\{t \geq \psi^{(i-1)} : R_t^{(i)} = \eta\right\} \text{ if } \{\cdots\} \neq \emptyset \text{ else } +\infty. \end{aligned}$$

Observe that a strong solution to $(R_t^{(i)}, \Theta_t^{(i)})_{t\in[\psi^{(i-1)},\psi^{(i)}[}$ exists by Remark 2.4 since the process never hits the lines d^1 (resp. d^{-1}) when $(-1)^{i-1} = 1$ (resp. $(-1)^{i-1} = -1$).

Let $\psi^{(0)} = 0$. The process $(A_t)_{t\in[0,+\infty[}$ is defined by $(A_t) = (-1)^{i-1}A_0$ on $[\psi^{(i-1)}, \psi^{(i)}[$. Define

$$\tilde{i} = \inf\{i > 0 : R^{(i)}_{\psi^{(i)}} = 0\} \text{ if } \{\cdots\} \neq \emptyset \text{ else } +\infty.$$

The event $\{\psi^{(\tilde{i})} < \infty\}$ means that the yacht reaches the origin in a finite amount of time. Observe that, on this event, for $i \geq \tilde{i}$, then $\psi^{(i)} = \psi^{(\tilde{i})}$, so after tacking at $\psi^{(\tilde{i}-1)}$, the yacht reaches the buoy.

The process $(R_t^A, \Theta_t^A)_{i\in[0,\infty[}$ is defined as

$$(R_t^A, \Theta_t^A) = \begin{cases} (R_t^{(i)}, \Theta_t^{(i)}) \text{ for } i \leq \tilde{i},\ t \in [\psi^{(i-1)}, \psi^{(i)}[\\ (0, \Theta^{(\tilde{i})}_{\psi^{(\tilde{i})}}) \text{ for } t \geq \psi^{(\tilde{i})}. \end{cases} \tag{4.1}$$

In order to show that A satisfies Item (a) of Definition 3.4 we need to prove that it is piecewise continuous and that (3.1) admits a strong solution for $t \in [0, \infty[$.

Lemma 4.1. *The process (A_t) defined above is piecewise constant. In particular there exists a strong solution to $(R_t^A, \Theta_t^A)_{t\in[0,+\infty[}$ defined in* (4.1).

Proof. We show that A is piecewise constant. Fix $\omega \in \Omega$ and $T < \infty$. In order for the strategy A_t to switch from ± 1 to ∓ 1, it is required at least an increment greater than α of the Brownian motion, indeed the yacht has to cross on port tack (resp. starboard tack) the interval $[\frac{3\pi}{2}, \frac{3\pi}{2} + \alpha]$ or $[-\alpha, 0]$ (resp. $[\frac{\pi}{2}, \frac{\pi}{2} + \alpha]$ or $[\pi - \alpha, \pi]$). In these intervals, $\mu_2(R_t, \Theta_t, -1) = 0$ (resp. $\mu_2(R_t, \Theta_t, 1) = 0$). By uniform continuity of the path of Brownian motion in $[0, T]$, for every ε, there exists $\delta : \forall |t_1 - t_2| \leq \delta, |B_{t_1} - B_{t_2}| \leq \varepsilon$. Let $\varepsilon = \alpha$. Then, for each $i \leq i^T := \inf\{i : \psi_i \geq T\}$, the holding times $\psi_i - \psi_{i-1}$ of the strategy are larger than δ and this proves that A is piecewise constant.

Since the strategy is piecewise constant and the process never hits the lines d^1 (resp. d^{-1}) when $A = 1$ (resp. $A = -1$), there exists a strong solution to (3.1) defined by (4.1). □

Remark 4.2. Now that the process $(R_t^A, \Theta_t^A)_{t\in[0,\infty[}$ is defined, it is straightforward to verify that $\tau_\eta^A = \nu^{(\tilde{i})}$ a.s., where the first is defined in (3.2). From now on, we will denote $\tau^A := \tau_\eta^A$.

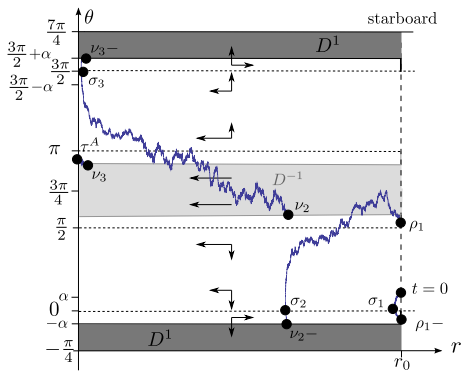

Figure 7: Tacking and continuation regions of the strategy A. The arrows indicate the direction and the intensity of the drift in the two directions r and θ. Each time the process hits D^1 (dark grey), A tacks. We picture this by a jump to the symmetric point on D^{-1} (light grey). Notice that D^1 contains the vertical segments $\{r_0\} \times \left[-\frac{\pi}{4}, 0\right]$ and $\{r_0\} \times \left[\frac{3\pi}{2}, \frac{7\pi}{4}\right[$.
The position of the yacht at the stopping times of (4.4) is depicted with bullet points.

Observe that the symmetry of the problem allows us to consider only one tack. Indeed, in Cartesian coordinates, it is clear that the payoff function (3.9) satisfies, for all $(x, y) \in \mathbb{R}^2$,

$$J^{c,\eta}((x, y), 1) = J^{c,\eta}((y, x), -1).$$

Likewise, in polar coordinates, we have, for all $(r, \theta) \in \mathbb{R}_+ \times \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[$,

$$J^{c,\eta}((r, \theta), 1) = J^{c,\eta}\left(\left(r, \frac{\pi}{2} - \theta\right), -1\right), \tag{4.2}$$

where $\frac{\pi}{2} - \theta$ is considered modulo 2π in $\left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[$. Therefore, instead of considering two maps, one for each tack, as in Fig.6, we can consider only one map corresponding to the starboard tack. When the strategy A makes the yacht tack, then the process (R_t^A, Θ_t^A) makes a jump in the same map so that $\Theta_t^A \in [-\alpha, \frac{3\pi}{2} + \alpha]$ for $t \geq 0$. Namely, if $(R_{t-}^A, \Theta_{t-}^A) \in \partial C^1$, then (see Fig. 7) $(R_t^A, \Theta_t^A) = (R_t^A, \frac{\pi}{2} - \Theta_{t-}^A)$. In particular, if $\Theta_{t-}^A = -\alpha$ (resp. $\Theta_{t-}^A = \frac{3\pi}{2} + \alpha$), then $\Theta_t^A = \frac{\pi}{2} + \alpha$ (resp $\Theta_t^A = \pi - \alpha$). From now on, we will consider that the yacht is always on starboard tack and as a result of which we shorten the notations. For instance, $\mathbb{P}_{r,\theta,1}$ becomes $\mathbb{P}_{r,\theta}$. Notice the other symmetry condition given by

$$J^{c,\eta}((r, \theta), 1) = J^{c,\eta}\left(\left(r, \frac{3\pi}{2} - \theta\right), 1\right), \qquad \forall\, (r, \theta) \in \mathbb{R}_+ \times \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[, \tag{4.3}$$

where $\frac{3\pi}{2} - \theta$ is considered modulo 2π in $\left[-\frac{\pi}{4}, \frac{7\pi}{4}\right[$.

We now define some stopping times which will be used in the proof of Lemmas 4.6, 4.8, 4.9. Recall that τ^A is defined in Remark 4.2.

Definition 4.3. Let (R_t^A, Θ_t^A) denote the position of a yacht driven by A. We set $\sigma_0 = \nu_0 = \rho_0 = \psi_0 = 0$, $E_0 = \emptyset$ and for all $i \in \mathbb{N}^*$, we define

$$\begin{aligned}
\sigma_i &= \inf\left\{t \geq \psi_{i-1} : \Theta_t^A \in \left[-\tfrac{\pi}{2}, 0\right]\right\} \wedge \tau^A \\
\nu_i &= \inf\left\{t \geq \psi_{i-1} : (R_t^A, \Theta_t^A) \in D_1^1\right\} \wedge \tau^A \\
\rho_i &= \inf\left\{t \geq \psi_{i-1} : (R_t^A, \Theta_t^A) \in D_2^1\right\} \wedge \tau^A \\
\psi_i &= \nu_i \wedge \rho_i \\
u_i &= \psi_i - \psi_{i-1}
\end{aligned} \tag{4.4}$$

and

$$E_i = \left\{ \sup_{\psi_{i-1} \le t \le \tau^A} \left| \sigma B_t - \sigma B_{\psi_{i-1}} \right| < \tfrac{\alpha}{2} \right\} \subset \Omega.$$

Observe that, once the process (R_t, Θ_t) is defined as in (4.1), then $\nu_i = \nu^{(i)}$, $\rho_i = \rho^{(i)}$, $\psi_i = \psi^{(i)}$ a.s.

The stopping times of (4.4) are illustrated in Fig.7. The times ψ_i are the tacking times of the strategy A and the event E_i means that after the $(i-1)$-th tacking, the wind is calm (i.e. varies by no more than $\frac{\alpha}{2}$) until the end of the race.

Remark 4.4. By definition, $\psi_i \le \tau^A$ and if $\{\psi_i = \tau^A\}$ occurs, then E_{i+1} occurs trivially.

Remark 4.5. Below Lemmas 4.6, 4.8, 4.9 and Corollary 4.10, 4.11 altogether tell us that, if from a tacking time ψ_{i-1}, the wind is calm for a sufficiently long time (that is E_i occurs), then the yacht will reach the buoy with at most one tacking and in a bounded time. Moreover, if the wind is calm, then the yacht will never be pushed backwards into the circle $r = r_0$.

Lemma 4.6. *There exists $K \in \mathbb{R}_+$ such that for all $i \in \mathbb{N}$ and for all $(r, \theta) \in \bar{C}^1$, the holding times u_i of the strategy A defined in* (4.4) *satisfy*

$$\mathbb{E}_{r,\theta}(u_i) \le K < +\infty.$$

Moreover, the stopping times σ_i, ν_i and ψ_i defined in (4.4) *are integrable.*

Proof. Let $(r, \theta) \in \bar{C}^1$ be given numbers and let $f(x) = \left(x - \frac{3\pi}{4}\right)^2$. Then by Itô's formula, we have

$$\begin{aligned} f\left(\Theta^A_{t\wedge\nu_1}\right) - f(\theta) = &\int_0^{t\wedge\nu_1} 2\mu_2\left(R^A_s, \Theta^A_s, 1\right)\left(\Theta^A_s - \tfrac{3\pi}{4}\right) ds \\ &+ 2\sigma \int_0^{t\wedge\nu_1} \left(\Theta^A_s - \tfrac{3\pi}{4}\right) dB_s + \sigma^2 (t \wedge \nu_1). \end{aligned}$$

On one hand, Θ^A_t is bounded. On the other hand, for all $(r, \theta) \in \bar{C}^1$, $f(\theta) \le \left(-\alpha - \frac{3\pi}{4}\right)^2$ and $\mu_2(r, \theta, 1)\left(\theta - \frac{3\pi}{4}\right) \ge 0$ (see (2.12)). Hence, taking the expectation in the previous equation and using these inequalities, we find that $\sigma^2 \mathbb{E}_{r,\theta}(t \wedge \nu_1) \le 2\left(\alpha + \frac{3\pi}{4}\right)^2$. Therefore, by the monotone convergence theorem, we find that

$$\mathbb{E}_{r,\theta}(\nu_1) \le 2\sigma^{-2}\left(\alpha + \tfrac{3\pi}{4}\right)^2 =: K.$$

Observe that by definition of these stopping times, $\mathbb{E}_{r,\theta}(u_1) = \mathbb{E}_{r,\theta}(\psi_1) = \mathbb{E}_{r,\theta}(\rho_1 \wedge \nu_1) \le \mathbb{E}_{r,\theta}(\nu_1)$. Furthermore, by the strong Markov property, we have for all $i \in \mathbb{N}$,

$$\mathbb{E}_{r,\theta}(u_i) = \mathbb{E}_{r,\theta}\left(\mathbb{E}_{r,\theta}\left(u_i \mid \mathcal{F}_{\psi_{i-1}}\right)\right) = \mathbb{E}_{r,\theta}\left(\mathbb{E}_{R^A_{\psi_{i-1}}, \Theta^A_{\psi_{i-1}}}(u_1)\right) \le K.$$

Finally, $\nu_i, \psi_i, \sigma_i \le \sum_{n=1}^{i} u_i$. Thus, all these stopping times are of finite expectation. □

Remark 4.7. A consequence of this lemma is that, without loss of generality, we can consider initial conditions (r, θ) such that $r \le r_0$ and $\theta \in \left]\frac{\pi}{2} - \alpha, \frac{\pi}{2} + \alpha\right]$. Indeed, with at most one tacking and in a time with finite expectation, using (4.2), the yacht will be at

$$(R^A_{\psi_1}, \Theta^A_{\psi_1}) \in \left([\eta, r_0[\times \left\{\tfrac{\pi}{2} + \alpha, \pi - \alpha\right\}\right) \cup \left(\{r_0\} \times \left(\left[\tfrac{\pi}{2}, \tfrac{\pi}{2} + \alpha\right] \cup [\pi - \alpha, \pi]\right)\right).$$

Moreover, by symmetry (see (4.3)), we can assume that

$$(R^A_{\psi_1}, \Theta^A_{\psi_1}) \in \left([\eta, r_0[\times \left\{\tfrac{\pi}{2}+\alpha\right\}\right) \cup \left(\{r_0\} \times \left[\tfrac{\pi}{2}, \tfrac{\pi}{2}+\alpha\right]\right)$$

and this set is clearly included in $[\eta, r_0] \times \left]\frac{\pi}{2}-\alpha, \frac{\pi}{2}+\alpha\right]$. From now on, we will denote

$$\mathcal{S} := [\eta, r_0] \times \left]\tfrac{\pi}{2}-\alpha, \tfrac{\pi}{2}+\alpha\right]. \tag{4.5}$$

Next two lemmas prove bounds on the stopping times introduced in (4.4). Moreover, they show that, if the wind is calm, that is E_1 occurs, then the yacht is never pushed backwards into the circle of radius r_0.

Lemma 4.8. *If ζ denotes the following stopping time*

$$\zeta = \inf\left\{t \geq 0 : \Theta^A_t \notin \left[\tfrac{\pi}{2}-\alpha,\ \tfrac{\pi}{2}+\alpha\right]\right\} \wedge \tau^A,$$

then, for all $(r, \theta) \in \mathcal{S}$,

$$R^A_t \leq r, \qquad t \in [0, \zeta] \qquad \textit{and} \qquad \zeta \leq \frac{r-\eta}{v\cos\alpha}, \qquad \mathbb{P}_{r,\theta}\textit{-almost surely:}$$

Proof. By (2.12), we have for all $0 \leq t \leq \zeta$ that $\mu_1\left(\Theta^A_t, 1\right) \in [-v, -v\cos\alpha]$. Hence,

$$\eta \leq R^A_{t\wedge\zeta} = r + \int_0^{t\wedge\zeta} \mu_1\left(\Theta^A_s, 1\right)\, ds \leq r - v\cos\alpha\,(t\wedge\zeta), \quad \forall\, t \geq 0.$$

Therefore, $R^A_t \leq r$, for $t \in [0, \zeta]$, and $\zeta \leq \frac{r-\eta}{v\cos\alpha}$. □

Lemma 4.9. *If the event E_1 occurs, then for all $(r, \theta) \in \mathcal{S}$, the following inequalities hold $\mathbb{P}_{r,\theta}$-almost surely.*

$$\sigma_1 - \zeta \leq \frac{d_1 r}{v}, \tag{4.6a}$$

$$\psi_1 - \sigma_1 \leq \frac{d_2 R^A_{\sigma_1}}{v}, \tag{4.6b}$$

where

$$d_1 = \sin\left(\tfrac{\alpha}{2}\right)\tan\alpha + 1, \qquad d_2 = \frac{\left(\sin\left(\frac{\alpha}{2}\right) + \tan\alpha\right)\cos\alpha}{\cos(2\alpha)}.$$

Moreover,

$$R^A_t \leq r, \qquad t \in [0, \psi_1]. \tag{4.7}$$

Proof. In order to establish this statement, we will treat three cases separately.

Case 1: $R^A_\zeta = \eta$ (which is equivalent to $\zeta = \tau^A$). If $\zeta = \tau^A$, then $M_{\tau^A}(A) = 0$, $\sigma_1 = \nu_1 = \rho_1 = \psi_1 = \tau^A$ and since $R^A_{\tau^A} = \eta$, the two inequalities are satisfied and (4.7) follows from Lemma 4.8.

Case 2: $\Theta^A_\zeta = \frac{\pi}{2} + \alpha$ and $R^A_\zeta > \eta$. Then, on the event E_1, we have, for all $t \geq \zeta$, that $\Theta^A_t \in \left[\frac{\pi}{2}, \pi\right]$. Indeed, if we set

$$\kappa = \begin{cases} \inf\left\{t \geq \zeta : \Theta^A_t \notin \left[\frac{\pi}{2}, \pi\right]\right\}, & \text{if } \{\cdots\} \neq \emptyset, \\ +\infty, & \text{otherwise,} \end{cases}$$

then, by (2.12) and (2.9), on $\{\kappa < +\infty\}$,

$$\Theta_\kappa^A = \Theta_\zeta^A + \int_\zeta^\kappa \underbrace{\mu_2\left(R_s^A, \Theta_s^A, 1\right)}_{=0,\text{ since } s<\kappa} ds + \underbrace{\sigma B_\kappa}_{|\cdot|<\frac{\alpha}{2}} - \underbrace{\sigma B_\zeta}_{|\cdot|<\frac{\alpha}{2}} > \tfrac{\pi}{2} + \alpha - 2\tfrac{\alpha}{2} = \tfrac{\pi}{2},$$

and similarly,

$$\Theta_\kappa^A < \tfrac{\pi}{2} + \alpha + 2\tfrac{\alpha}{2} = \tfrac{\pi}{2} + 2\alpha < \pi.$$

Since $\Theta_\kappa^A \in [\frac{\pi}{2}, \pi]$ on $\{\kappa < +\infty\}$, this shows that $\mathbb{P}\{\kappa < +\infty\} = 0$ and therefore, $\kappa \equiv +\infty$. It follows that $\Theta_t^A \in \left[\frac{\pi}{2}, \pi\right]$ for all $t \geq \zeta$. We have then by (2.12) that for all $t \geq \zeta$,

$$R_{t\wedge\tau^A}^A = R_\zeta^A + \int_\zeta^{t\wedge\tau^A} \underbrace{\mu_1\left(\Theta_s^A, 1\right)}_{=-v} ds = R_\zeta^A - v(t\wedge\tau^A - \zeta). \tag{4.8}$$

Taking the limit when $t \to +\infty$, we find that $\tau^A - \zeta \leq \frac{R_\zeta^A - \eta}{v} \leq \frac{r}{v}$ by Lemma 4.8. Furthermore, since $\kappa = +\infty$, we have $\psi_1 = \sigma_1 = \nu_1 = \rho_1 = \tau^A$ and $R_{\sigma_1}^A = \eta \leq r$. Thus, the two inequalities (4.6a),(4.6b) are also satisfied in this case. By (4.8), it follows that $R_t^A \leq R_\zeta^A \leq r$ for $t \in [\zeta, \psi_1]$, where the last inequality follows from Lemma 4.8, hence also (4.7) is proved.

Case 3: $\Theta_\zeta^A = \frac{\pi}{2} - \alpha$ and $R_\zeta^A > \eta$. On E_1, we can show similarly as in the previous case that

$$\Theta_t^A \in \left[-\alpha, \tfrac{\pi}{2}\right[, \qquad \text{for all } t \in [\zeta, \psi_1[. \tag{4.9}$$

Therefore, by Prop 2.1 and by (2.2), since the yacht is in zone $I \cup IV^a$, in Cartesian coordinates the process $\vec{X}_t^A$ satisfies for all $t \in [\zeta, \psi_1[$

$$X_t^A = \cos(\sigma B_t) X_\zeta^A - \sin(\sigma B_t) Y_\zeta^A + v\int_\zeta^t \sin(\sigma B_t - \sigma B_s) ds,$$
$$Y_t^A = \sin(\sigma B_t) X_\zeta^A + \cos(\sigma B_t) Y_\zeta^A - v\int_\zeta^t \cos(\sigma B_t - \sigma B_s) ds.$$

On E_1, $|\sigma B_t| < \frac{\alpha}{2}$ for all $t \leq \tau^A$, thus, we can bound Y_t^A in the second equation as follows

$$-\sin\left(\tfrac{\alpha}{2}\right) X_\zeta^A + \cos\left(\tfrac{\alpha}{2}\right) Y_\zeta^A - v(t-\zeta) < Y_t^A < \sin\left(\tfrac{\alpha}{2}\right) X_\zeta^A + Y_\zeta^A - v(t-\zeta)\cos\alpha,$$

for all $t \in [\zeta, \psi_1[$. Replacing $X_\zeta^A = R_\zeta^A \cos\left(\frac{\pi}{2} - \alpha\right) = R_\zeta^A \sin\alpha$ and $Y_\zeta^A = R_\zeta^A \cos\alpha$ in the above equation, we get

$$\cos\left(\tfrac{3\alpha}{2}\right) R_\zeta^A - v(t-\zeta) < Y_t^A < \left(\sin\left(\tfrac{\alpha}{2}\right)\sin\alpha + \cos\alpha\right) R_\zeta^A - v(t-\zeta)\cos\alpha. \tag{4.10}$$

Observe that, on E_1 and under our assumptions, $Y_{\sigma_1}^A = C \geq 0$, for a costant C depending on $\omega \in \Omega$. Strict inequality corresponds to the yacht hitting the buoy before the line $y = 0$, that is $\sigma_1 = \tau^A$. Setting $t = \sigma_1$ and adding $v(t-\zeta)\cos\alpha$ to (4.10), the second inequality becomes

$$v(\sigma_1 - \zeta) < d_1 R_\zeta^A - \frac{C}{\cos\alpha}. \tag{4.11}$$

We have thus, by Lemma 4.8, that $\sigma_1 - \zeta < \frac{d_1 r}{v}$ and this establishes inequality (4.6a) of the statement. Notice that if $\sigma_1 = \tau^A$, then $\psi_1 = \nu_1 = \rho_1 = \tau^A$, $R_{\sigma_1}^A = \eta$ and (4.6b) is trivial, so from now on, we assume $\sigma_1 < \tau^A$.

On E_1, since $X^A_{\sigma_1} = R^A_{\sigma_1}$ and $Y^A_{\sigma_1} = 0$, we have for all $t \in [\sigma_1, \psi_1[$, by Prop 2.1 and by (2.2), since the yacht is in zone IV^a,

$$X^A_t = \cos(\sigma B_t) R^A_{\sigma_1} + v \int_{\sigma_1}^t \sin(\sigma B_t - \sigma B_s) ds,$$
$$Y^A_t = \sin(\sigma B_t) R^A_{\sigma_1} - v \int_{\sigma_1}^t \cos(\sigma B_t - \sigma B_s) ds,$$

and so $X^A_t \leq R^A_{\sigma_1} + v(t - \sigma_1) \sin\alpha$ and

$$-\sin\left(\tfrac{\alpha}{2}\right) R^A_{\sigma_1} - v(t-\sigma_1) \leq \; Y^A_t \; \leq \sin\left(\tfrac{\alpha}{2}\right) R^A_{\sigma_1} - v(t-\sigma_1)\cos\alpha. \tag{4.12}$$

Since on this time interval, $X^A_t \tan\alpha + Y^A_t > 0$, we find by combining these inequalities that

$$0 < X^A_t \tan\alpha + Y^A_t \leq R^A_{\sigma_1} \left(\tan\alpha + \sin\left(\tfrac{\alpha}{2}\right)\right) + v(t-\sigma_1)\left(\tan\alpha \sin\alpha - \cos\alpha\right),$$

which provides

$$t - \sigma_1 \leq \frac{R^A_{\sigma_1}}{v} \frac{\tan\alpha + \sin\left(\frac{\alpha}{2}\right)}{\cos\alpha - \tan\alpha\sin\alpha} = \frac{R^A_{\sigma_1}}{v} d_2 \tag{4.13}$$

by definition of d_2. Taking the limit $t \to \psi_1$, we obtain inequality (4.6b) of the statement.

We now prove inequality (4.7). By (4.9) and (2.12), on E_1, for all $t \in [\zeta, \psi_1[$, R^A_t satisfies

$$dR^A_t = -v \sin\left(\Theta^A_t\right) \, dt = -v \frac{Y^A_t}{R^A_t} \, dt. \tag{4.14}$$

Hence, we have

$$d\left(R^A_t\right)^2 = 2R^A_t \, dR^A_t = -2vY^A_t \, dt, \tag{4.15}$$

and by (4.10),

$$\begin{aligned}\left(R^A_t\right)^2 &\leq \left(R^A_\zeta\right)^2 - 2v \int_\zeta^t \left(\cos\left(\tfrac{3\alpha}{2}\right) R^A_\zeta - v(s-\zeta)\right) ds \\ &= \left(R^A_\zeta\right)^2 + \left(v(t-\zeta) - \cos\left(\tfrac{3\alpha}{2}\right) R^A_\zeta\right)^2 - \cos^2\left(\tfrac{3\alpha}{2}\right) \left(R^A_\zeta\right)^2.\end{aligned}$$

Setting $t = \sigma_1$ and using (4.11), we get by Lemma 4.8

$$\left(R^A_{\sigma_1}\right)^2 \leq \left(R^A_\zeta\right)^2 + \left(d_1 R^A_\zeta - \cos\left(\tfrac{3\alpha}{2}\right) R^A_\zeta\right)^2 - \cos^2\left(\tfrac{3\alpha}{2}\right) \left(R^A_\zeta\right)^2 \leq r^2 c_1^2, \tag{4.16}$$

where $c_1 = \sqrt{1 + \left(d_1 - \cos\left(\frac{3\alpha}{2}\right)\right)^2 - \cos^2\left(\frac{3\alpha}{2}\right)}$. Moreover, by (4.15), we have that on E_1, for all $t \in [\sigma_1, \psi_1[$

$$\begin{aligned}\left(R^A_t\right)^2 &= \left(R^A_{\sigma_1}\right)^2 - 2v \int_{\sigma_1}^t Y^A_s \, ds \\ \text{(by (4.12))} \qquad &\leq \left(R^A_{\sigma_1}\right)^2 + 2v \int_{\sigma_1}^t \left(\sin\left(\tfrac{\alpha}{2}\right) R^A_{\sigma_1} + v(s-\sigma_1)\right) ds \\ &= \left(R^A_{\sigma_1}\right)^2 + \left(\sin\left(\tfrac{\alpha}{2}\right) R^A_{\sigma_1} + v(t-\sigma_1)\right)^2 - \sin^2\left(\tfrac{\alpha}{2}\right) \left(R^A_{\sigma_1}\right)^2 \\ \text{(by (4.13))} \qquad &\leq \left(R^A_{\sigma_1}\right)^2 \left(1 + \left(\sin\left(\tfrac{\alpha}{2}\right) + d_2\right)^2 - \sin^2\left(\tfrac{\alpha}{2}\right)\right) = \left(R^A_{\sigma_1}\right)^2 c_2^2,\end{aligned} \tag{4.17}$$

where $c_2 = \sqrt{1 + \left(d_2 + \sin\left(\frac{\alpha}{2}\right)\right)^2 - \sin^2\left(\frac{\alpha}{2}\right)}$. Since the process R_t^A is continuous, this inequality remains true for $t = \psi_1$.

The last assertion (4.7) is true for $t \in [0, \zeta]$ by Lemma 4.8. For $t \in [\zeta, \sigma_1]$, (4.7) follows from (4.14) since $\sin(\Theta_t^A) > 0$. Finally, for $t \in [\sigma_1, \psi_1]$, (4.16) and (4.17) imply that $R_t^A \leq c_1 c_2 r$. Long but direct computations show that

$$c_1^2 c_2^2 = \frac{1}{2}\left(1 + \cos\alpha + 2\left((1 + \cos\alpha \sec(2\alpha)) \sin\left(\tfrac{\alpha}{2}\right) + \sec(2\alpha)\sin(\alpha)\right)^2\right) \\ \times \left(1 - \cos\left(\tfrac{3\alpha}{2}\right)^2 + \left(1 - \cos\left(\tfrac{3\alpha}{2}\right) + \sin\left(\tfrac{\alpha}{2}\right)\tan\alpha\right)^2\right).$$

where $\sec = \frac{1}{\cos}$. Furthermore, it can be shown that the term on the right-hand side is an increasing and positive function of $\alpha \in \left[0, \frac{\pi}{4}\right[$ approximately equal to 0.72 at $\frac{\pi}{8}$. Since we assumed that $\alpha \leq \frac{\pi}{8}$, we conclude that $c_1 c_2 < 1$ and this concludes the proof. □

The Corollary below proves that, if at some point of the race the wind is calm, then the yacht reaches the origin with at most one additional tack. We recall that the region $\mathcal{S}$ is defined in (4.5).

Corollary 4.10. *For all $(r, \theta) \in \mathcal{S}$ and $i \in \mathbb{N}$, if E_i occurs then $\nu_i = \psi_i \leq \tau^A = \psi_{i+1}$ and $M_{\tau^A}(A) \leq i$ $\mathbb{P}_{r,\theta}$-a.s. Moreover, if $\psi_i < \tau^A$, then $\Theta_{\psi_i}^A = \frac{\pi}{2} + \alpha$.*

Proof. By construction, at time ψ_{i-1}, the yacht is in $\mathcal{S}$. If $\psi_i = \tau^A$ the proof is trivial, so we assume $\psi_i < \tau^A$, hence we are in the situation described by case 3 of the proof of Lemma 4.9. By (4.7) and by the strong Markov property, we have that $R_t^A < r_0$, $\mathbb{P}_{r,\theta}$-a.s. for all $t \in [\psi_{i-1}, \psi_i]$ on E_i. Thus, at time ψ_i- the yacht hits the line $\theta = -\alpha$, $\nu_i < \rho_i$ and $\Theta_{\psi_i}^A = \frac{\pi}{2} + \alpha$. Again by the strong Markov property, this configuration is equivalent to the case 2 of the proof of Lemma 4.9. Therefore, on E_i, in this case we have that $\sigma_{i+1} = \nu_{i+1} = \psi_{i+1} = \tau^A$ and so $M_{\tau^A}(A) \leq i$. □

Corollary 4.11. *If E_1 occurs then, for all $(r, \theta) \in \mathcal{S}$,*

$$\tau^A < (\sec\alpha + 1)\frac{r_0 - \eta}{v} + (d_1 + d_2)\frac{r_0}{v} =: \Gamma, \qquad \mathbb{P}_{r,\theta}\text{-almost surely.}$$

Proof. Indeed, on E_1, we have by the preceding corollary that $M_{\tau^A}(A) \leq 1$. If $\psi_1 < \tau^A$, then $\nu_1 = \psi_1$ and $\Theta_{\psi_1}^A = \frac{\pi}{2} + \alpha$. Thus, by Lemma 4.8 and Lemma 4.9,

$$\tau^A = \zeta + (\sigma_1 - \zeta) + (\psi_1 - \sigma_1) + (\tau^A - \psi_1) \leq \frac{r_0 - \eta}{v\cos\alpha} + \frac{d_1 r_0}{v} + \frac{d_2 r_0}{v} + (\tau^A - \psi_1),\ \mathbb{P}_{r,\theta}\text{-a.s..}$$

Furthermore, on E_1, $\Theta_t^A \in \left[\frac{\pi}{2}, \pi\right]$ for all $t \in [\psi_1, \tau^A]$, hence $R_t^A = R_{\psi_1}^A - v(t - \psi_1)$ and

$$\tau^A - \psi_1 = \frac{R_{\psi_1}^A - R_{\tau^A}^A}{v} = \frac{R_{\psi_1}^A - \eta}{v} < \frac{r_0 - \eta}{v}.$$

The case $\psi_1 = \tau^A$ follows from above. □

Proposition 4.12. *There exists $p_0 \in]0, 1[$, depending only on the parameters α, σ, v and r_0, such that for all $(r, \theta) \in \mathcal{S}$,*

(a) $\mathbb{P}_{r,\theta}\left(\psi_2 = \tau^A\right) \geq p_0$,

(b) $\mathbb{P}_{r,\theta}\left(\psi_i < \tau^A\right) \leq (1 - p_0)^{\lfloor \frac{i}{2} \rfloor}$ *for all $i \in \mathbb{N}$, $i \geq 2$.*

Proof. **(a)** Observe that by Corollary 4.10, $E_1 \subset \left\{\psi_2 = \tau^A\right\}$. Thus, by Corollary 4.11,

$$\begin{aligned}\mathbb{P}_{r,\theta}\left(\psi_2 = \tau^A\right) \geq \mathbb{P}_{r,\theta}(E_1) &= \mathbb{P}_{r,\theta}\left(\sup_{0\leq t\leq \tau^A} |\sigma B_t| < \tfrac{\alpha}{2}\right) \\ &\geq \mathbb{P}_{r,\theta}\left(\sup_{0\leq t\leq \Gamma} |\sigma B_t| < \tfrac{\alpha}{2}\right) =: p_0 > 0.\end{aligned}$$

(b) Notice that $\left\{\psi_i < \tau^A\right\} \subset \left\{\psi_{i-1} < \tau^A\right\} \subset \cdots \subset \left\{\psi_1 < \tau^A\right\}$, so by the strong Markov property and the previous point, we get

$$\begin{aligned}\mathbb{P}_{r,\theta}\left(\psi_i < \tau^A\right) &= \mathbb{P}_{r,\theta}\left(\left\{\psi_i < \tau^A\right\} \cap \left\{\psi_{i-2} < \tau^A\right\}\right) \\ &= \mathbb{E}_{r,\theta}\left(\mathbf{1}_{\{\psi_{i-2}<\tau^A\}}\, \mathbb{P}_{r,\theta}\left(\psi_i < \tau^A \mid \mathcal{F}_{\psi_{i-2}}\right)\right) \\ &\leq \mathbb{P}_{r,\theta}(\psi_{i-2} < \tau^A)(1-p_0).\end{aligned}$$

If i is even, the induction is stopped at $\mathbb{P}_{r,\theta}(\psi_2 < \tau^A)$ and claim follows. If i is odd, the induction is stopped at $\mathbb{P}_{r,\theta}(\psi_3 < \tau^A)$ and the conclusion follows after remarking that $\mathbb{P}_{r,\theta}(\psi_3 < \tau^A) \leq \mathbb{P}_{r,\theta}(\psi_2 < \tau^A)$. □

Proposition 4.13. *For all $(r,\theta) \in \mathcal{S}$,*

(a) $\mathbb{E}_{r,\theta}\left(M_{\tau^A}(A)\right) < \frac{2}{p_0} - 1$,

(b) $\mathbb{E}_{r,\theta}\left(\tau^A\right) < K\frac{2}{p_0}$,

where K is the constant defined in Lemma 4.6.

Proof. **(a)** Notice that $\{M_{\tau^A}(A) > i\} \subset \{\psi_{i+1} < \tau^A\}$, thus, we have, by Prop. 4.12

$$\begin{aligned}\mathbb{E}_{r,\theta}\left(M_{\tau^A}(A)\right) &= \sum_{i=0}^{\infty} \mathbb{P}_{r,\theta}\left(M_{\tau^A}(A) > i\right) \leq \sum_{i=0}^{\infty} \mathbb{P}_{r,\theta}\left(\psi_{i+1} < \tau^A\right) \\ &\leq 1 + \sum_{i=2}^{\infty} \mathbb{P}_{r,\theta}\left(\psi_i < \tau^A\right) \leq 1 + \sum_{i=2}^{\infty} (1-p_0)^{\lfloor \frac{i}{2} \rfloor} = \frac{2}{p_0} - 1.\end{aligned}$$

(b) We have

$$\begin{aligned}\mathbb{E}_{r,\theta}\left(\tau^A\right) &= \mathbb{E}_{r,\theta}\left(\sum_{i=0}^{M_{\tau^A}(A)} u_{i+1}\right) = \mathbb{E}_{r,\theta}\left(\sum_{k=0}^{\infty}\sum_{i=0}^{k} u_{i+1}\mathbf{1}_{\left\{M_{\tau^A}(A)=k\right\}}\right) \\ &= \sum_{i=0}^{\infty} \mathbb{E}_{r,\theta}\left(\sum_{k=i}^{\infty} u_{i+1}\mathbf{1}_{\left\{M_{\tau^A}(A)=k\right\}}\right) = \sum_{i=0}^{\infty} \mathbb{E}_{r,\theta}\left(u_{i+1}\mathbf{1}_{\left\{M_{\tau^A}(A)\geq i\right\}}\right) \\ &\leq \sum_{i=0}^{\infty} \mathbb{E}_{r,\theta}\left(\mathbb{E}_{r,\theta}\left(u_{i+1} \mid \mathcal{F}_{\psi_i}\right)\mathbf{1}_{\{\psi_i<\tau^A\}}\right).\end{aligned}$$

Then, by the strong Markov property and by Lemma 4.6, we find that the right-hand side is smaller than

$$K\sum_{i=0}^{\infty} \mathbb{P}_{r,\theta}\left\{\psi_i < \tau^A\right\} \leq K\left(2 + \sum_{i=2}^{\infty}(1-p_0)^{\lfloor \frac{i}{2} \rfloor}\right) = K\frac{2}{p_0},$$

and the proof of Item (b) is complete. □

The Theorem below is the main result of the section. As a side result, it also proves an upper bound on the Value function (3.4), for the case $c = 0$, which will be improved in the next section, where the optimal strategy is conjectured. We recall that $V^{c,\eta}(r,\theta,a)$ (resp. $V^{\eta}(r,\theta)$) is defined in (3.10) (resp. (3.4)).

Theorem 4.14. *For every $\eta > 0$, $c \geq 0$ and every $(r,\theta,a) \in \mathbb{R}_+ \times \mathbb{R} \times \{\pm 1\}$, $A \in \mathcal{A}^{\eta}_{r,\theta} \cap \mathcal{A}^{c,\eta}_{r,\theta,a}$. In particular,*

(a) $V^{c,\eta}(r,\theta,a) \leq K(1+\frac{2}{p_0}) + c\left(1+\frac{2}{p_0}\right)$,

(b) $V^{\eta}(r,\theta) \leq K(1+\frac{2}{p_0})$,

where K is the constant defined in Lemma 4.6.

Proof. Let $r_0 > r$. Proposition 4.13 shows that the strategy A satisfies

$$J^{c,\eta}(r,\theta,a,A) \leq K\frac{2}{p_0} + c\left(\frac{2}{p_0} - 1\right),$$

for all $(r,\theta) \in \mathcal{S}$ and $a = 1$. Remark 4.7 tells us that if the initial conditions are different, i.e. $a = 1$ and $(r,\theta) \in C^1$, it is a matter of one tacking and of a finite expectation time to be in the initial configuration, so $J^{c,\eta}(r,\theta,a,A) \leq K(1+\frac{2}{p_0}) + c\left(\frac{2}{p_0}\right)$. If $a = -1$ and $(r,\theta) \in C^{-1}$, then symmetry (4.2) tells us that it is equivalent to consider $a = 1$ and $(r, \frac{\pi}{2} - \theta)$, which is in C^1. Finally, if $a = 1$ and $(r,\theta) \in D^1$ (resp. $a = -1$ and $(r,\theta) \in D^{-1}$), the yacht tacks immediately, so $J^{c,\eta}(r,\theta,a,A) \leq K(1+\frac{2}{p_0}) + c\left(1+\frac{2}{p_0}\right)$. This proves Item a) and together with Lemma 4.1, shows that $A \in \mathcal{A}^{c,\eta}_{r,\theta,a}$ for all $\eta > 0$, $(r,\theta) \in \mathbb{R}_+ \times \mathbb{R}$, $a \in \{+1,-1\}$. Using Remark 3.5, it also follows that $A \in \mathcal{A}^{\eta}_{r,\theta}$. Item (b) follows from Item a) after discarding the tacking cost. □

5 Conjectured optimal strategy for $\eta > 0$ and $c = 0$

The strategy studied in the previous section, while it allowed to prove that the Value function is finite in both cases $c > 0$ and $c = 0$, is suboptimal for the latter one. Indeed, for example, if $\Theta_t \in [0, \frac{\pi}{4}[$, depending of the path of the boat, there is a positive probability that $A_t = 1$. The boat, in this event, would proceed at speed $v\sin(\Theta_t)$ towards the buoy, while if $A_t = -1$ the speed would be $v\cos(\Theta_t)$.

We hypothesize that the optimal strategy, in the case of null tacking cost, involves a feedback approach, selecting, at any given position of the race field, the tack that maximizes the projection of the boat's speed along the radial direction. We prove that it is an admissible strategy in the sense of Definition 3.1. The coefficients of the state equation are not Lipschitz, hence standard results cannot be applied, but they exhibit a monotonicity property which allows to prove strong uniqueness and apply the Yamada-Watanabe argument. In a companion paper it will be shown that it is indeed optimal.

5.1 Definition of the strategy

Let $\eta > 0$ and (r,θ) be the initial fixed starting point. The conjectured optimal strategy has the following form.

Let $A^{\star}_{0-}(\theta) = a$ for a given $a \in \{-1,+1\}$. We recall that, since we are in the setting $c = 0$, the initial tack has no influence. Then for $t \geq 0$, $A^{\star}$ is defined as the feedback strategy

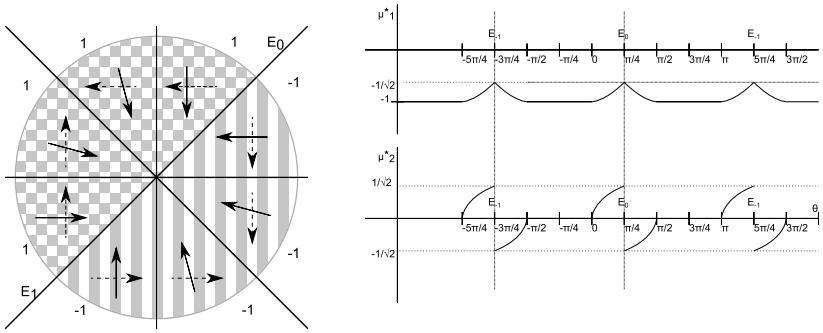

Figure 8: On the left, guessed optimal strategy $A^\star$. The arrows indicate the direction of the optimal drift, and the dashed ones the direction of the drift under the strategy $-A^\star$. On the right, drift of the processes $R^{A^\star}, \Theta^{A^\star}$ extended to $\mathbb{R}$ by 2π-periodicity.

$A^\star_t(\omega) := a^\star(\Theta_t(\omega))$, where

$$a^\star(\theta) = \begin{cases} 1, & \theta \in [\frac{\pi}{4}, \frac{5\pi}{4}[, \\ -1, & \theta \in [\frac{-3\pi}{4}, \frac{\pi}{4}[, \end{cases}$$

and then extended on $\mathbb{R}$ by 2π-periodicity. See also Fig. 8 for a visual interpretation. Under the strategy $A^\star$, system (3.1) becomes

$$\begin{cases} dR^{A^\star}_t &= 1_{\{R^{A^\star}_t > 0\}} \mu^\star_1(\Theta^{A^\star}_t) dt, \\ d\Theta^{A^\star}_t &= 1_{\{R^{A^\star}_t > 0\}} \mu^\star_2(R^{A^\star}_t, \Theta^{A^\star}_t) dt + \sigma 1_{\{R^{A^\star}_t > 0\}} dB_t, \end{cases} \tag{5.1}$$

where, since for $\theta \in [\frac{\pi}{4}, \frac{5\pi}{4}[$ the yacht sails on starboard tack,

$$\mu^\star_1(\theta) := \mu_1(\theta, 1), \qquad \mu^\star_2(r, \theta) := \mu_2(r, \theta, 1), \tag{5.2}$$

and for $\theta \in [-\frac{3\pi}{4}, \frac{\pi}{4}[$, the yacht is on port tack, therefore

$$\mu^\star_1(\theta) := \mu_1(\theta, -1), \qquad \mu^\star_2(r, \theta) := \mu_2(r, \theta, -1). \tag{5.3}$$

The functions $\mu^\star_1$, $\mu^\star_2$ are then extended to $\theta \in \mathbb{R}$ by 2π-periodicity. We recall the definition of μ_1 and μ_2 in (2.13).

5.2 Existence of the process controlled by $A^\star$, case $\eta \geq 0$

Proving the existence and uniqueness of a solution of the system (5.4) controlled by the process $A^\star$ requires some care, because the function $\theta_0 \to \mu_2(r, \theta)$ has discontinuities at $G := \cup_{k \in \mathbb{Z}} G_k$, where $G_k = \{\frac{\pi}{4} + k\pi\}$, therefore standard results do not apply. In the present situation the coefficients satisfy a monotonicity property, see (5.5), but even the results in literature which rely on this property assume continuity of the coefficients in the spatial coordinates, and hence are not applicable, see for example [7, Sec. 3]. Our proof uses the classical Yamada-Watanabe procedure, therefore we first prove pathwise uniqueness, and then the existence of a weak solution.

Proposition 5.1. *Let $\eta > 0$. For every $(r, \theta) \in \mathbb{R}_+ \times \mathbb{R}$ there exists a unique strong solution to (5.1) on $[0, T]$, for any $T > 0$. Moreover $\tau^{A^\star}_\eta \leq \sqrt{2}\frac{(r-\eta)}{v}$ a.s. In particular, $A^\star$ is an admissible strategy, that is $A^\star \in \mathcal{A}^\eta$.*

Proof. Observe that, at or before an arbitrary time $T > 0$, the boat can be close to the origin, where the coefficient $\mu_2^\star$ blows up. For this reason, we first introduce the state equation below with damped coefficients. Define, for $n \geq n^\eta := \inf\{n > 0 : \frac{1}{n} < \eta\}$,

$$\phi_n(x) := \begin{cases} 0, & \text{if } x \leq \eta - \frac{1}{n}, \\ n(x-\eta)+1, & \text{if } x \in]\eta - \frac{1}{n}, \eta[, \\ 1, & \text{if } x \geq \eta, \end{cases}$$

and consider the system of SDEs

$$\begin{cases} dR_t^{n,A^\star} = \phi_n(R_t^{n,A^\star})\mu_1^\star(\Theta_t^{n,A^\star})dt, \\ d\Theta_t^{n,A^\star} = \phi_n(R_t^{n,A^\star})\mu_2^\star(R_t^{n,A^\star}, \Theta_t^{n,A^\star})dt + \phi_n(R_t^{n,A^\star})\sigma dB_t. \end{cases} \tag{5.4}$$

Define also the stopping time $\tau_\eta^{n,A^\star} := \inf\left\{t : R_t^{n,A^\star} \leq \eta\right\}$. Observe that $\tau_\eta^{n,A^\star}$ does not depend on n even if the process involved in its definition does. This is due to the fact that $\phi_n(r) = 1$ for $r \geq \eta$. Therefore, if there exists a unique strong solution to (5.4) for $t \in [0,T]$ and any $T > 0$, and in addition $\tau^{n,A^\star}$ is bounded, then $\tau_\eta^{A^\star} = \tau_\eta^{n,A^\star}$ a.s.

We first prove the existence of a strong solution to (5.4) for $t \in [0,T]$, $n \geq n^\eta$. Then, by a standard localization procedure, the existence and uniqueness of a strong solution to (5.1) follows for $t \in [0, \tau_\eta^{A^\star}]$ and $\mathbb{P}\left[(R_t^{A^\star}, T_t^{A^\star}) = (R_t^{n,A^\star}, \Theta_t^{n,A^\star}),\ t \in [0, \tau_\eta^{A^\star}]\right] = 1$. In the sequel, we will write $\mu_1(\theta)$ for $\mu_1(\theta)^\star$ and $\mu_2(r,\theta)$ for $\mu_2(r,\theta)^\star$.

The first step is to prove that μ_1 and μ_2 satisfy the monotonicity condition below

$$\begin{aligned} &\left(\begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} - \begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix}\right) \cdot \left(\begin{pmatrix} \phi_n(r_1)\mu_1(\theta_1) \\ \phi_n(r_1)\mu_2(r_1,\theta_1) \end{pmatrix} - \begin{pmatrix} \phi_n(r_2)\mu_1(\theta_2) \\ \phi_n(r_2)\mu_2(r_2,\theta_2) \end{pmatrix}\right) \\ &\quad \leq C_n[(r_1-r_2)^2 + (\theta_1-\theta_2)^2] \end{aligned} \tag{5.5}$$

where C_n is a constant depending on n. In fact, (5.5) can be rewritten as

$$(r_1-r_2)(\phi_n(r_1)\mu_1(\theta_1) - \phi_n(r_2)\mu_1(\theta_2)) + (\theta_1-\theta_2)(\phi_n(r_1)\mu_2(r_1,\theta_1) - \phi_n(r_2)\mu_2(r_2,\theta_2)). \tag{5.6}$$

After some straightforward calculations, using the fact that $(r,\theta) \to \phi_n(r)\mu_1(\theta)$ is Lipschitz on $\mathbb{R}^2$, and $r \to \phi_n(r)\mu_2(r,\theta)$ is Lipschitz (with Lipschitz constant that does not depend on θ), it is possible to show that (5.6) is smaller or equal than

$$C_n\mu_1(\theta_1)(r_1-r_2)^2 + C_n|r_1-r_2||\theta_1-\theta_2| + \phi_n(r_2)(\theta_1-\theta_2)(\mu_2(r_1,\theta_1) - \mu_2(r_2,\theta_2)), \tag{5.7}$$

where C_n is a positive constant depending on n which may vary from one occurrence to the other. The last term of (5.7) is

$$\begin{aligned} &\phi_n(r_2)(\theta_1-\theta_2)(\mu_2(r_1,\theta_1) - \mu_2(r_2,\theta_2)) \\ &\quad \leq \phi_n(r_2)(\theta_1-\theta_2)(\mu_2(r_1,\theta_1) - \mu_2(r_2,\theta_1) + \mu_2(r_2,\theta_1) - \mu_2(r_2,\theta_2)) \\ &\quad \leq C_n|\theta_1-\theta_2||r_1-r_2| + (\theta_1-\theta_2)(\mu_2(r_2,\theta_1) - \mu_2(r_2,\theta_2)). \end{aligned}$$

Recall the elementary inequality $ab \leq \frac{1}{2}(a^2+b^2)$. Therefore in order to get the desired inequality (5.5), it is enough to prove that for fixed r, there exists a positive constant C_n such that

$$(\theta_1-\theta_2)(\mu_2(r,\theta_1) - \mu_2(r,\theta_2)) \leq C_n(\theta_1-\theta_2)^2. \tag{5.8}$$

Let $F(\theta_1,\theta_2) := (\theta_1-\theta_2)(\mu_2(r,\theta_1) - \mu_2(r,\theta_2))$. First observe that $F(\theta_1,\theta_2) = F(\theta_2,\theta_1)$, so it is enough to consider the case $\theta_1 \leq \theta_2$.

We claim that it is enough to prove inequality (5.8) for θ_1 close to θ_2, say $|\theta_1 - \theta_2| \leq \frac{\pi}{4}$. Indeed assume that for a $\delta > 0$, (5.8) holds for $|\theta_1 - \theta_2| \leq \delta$. Let $M := \sup_{\theta \in \mathbb{R}} |\mu_2(r, \theta)|$, and set $D_n := \frac{2M}{\delta} \vee C_n$. This entails that, for $|\theta_1 - \theta_2| \geq \delta$

$$(\theta_1 - \theta_2)(\mu_2(r, \theta_1) - \mu_2(r, \theta_2)) \leq |\theta_1 - \theta_2| 2M \leq |\theta_1 - \theta_2| \delta D_n \leq D_n(\theta_1 - \theta_2)^2,$$

and if $|\theta_1 - \theta_2| < \delta$, then trivially,

$$(\theta_1 - \theta_2)(\mu_2(r, \theta_1) - \mu_2(r, \theta_2)) \leq D_n(\theta_1 - \theta_2)^2.$$

Now, it suffices to replace C_n by D_n in (5.8).

If $G_k \notin]\theta_1, \theta_2[$, for any $k \in \mathbb{Z}$, then $\theta \to \mu_2(r, \theta)$ is Lipschitz and (5.8) follows. Now let θ_1, θ_2 be such that for some $k \in \mathbb{Z}$, $G_k \in]\theta_1, \theta_2[$. Since the function $\theta \to \mu_2(r, \theta)$ is periodic, it suffices to consider:

(1) angles θ_1, θ_2 such that $G_1 = \frac{5\pi}{4} \in]\theta_1, \theta_2[$, and $|\theta_1 - \theta_2| \leq \frac{\pi}{4}$. This corresponds to $\theta_1 \in]\pi, \frac{5\pi}{4}[$ and $\theta_2 \in]\frac{5\pi}{4}, \frac{3\pi}{2}[$. The right hand-side of (5.8) becomes $v(\theta_1 - \theta_2)\left(-\frac{\sin\theta_1}{r} - \frac{\cos\theta_2}{r}\right) \leq 0$, hence (5.8) is trivially satisfied.

(2) angles θ_1, θ_2 such that $G_0 = \frac{\pi}{4} \in]\theta_1, \theta_2[$, and $|\theta_1 - \theta_2| \leq \frac{\pi}{4}$. This corresponds to $\theta_1 \in]0, \frac{\pi}{4}[$ and $\theta_2 \in]\frac{\pi}{4}, \frac{\pi}{2}[$. Similarly, the right hand-side of (5.8) becomes $v(\theta_1 - \theta_2)\left(\frac{\sin\theta_1}{r} + \frac{\cos\theta_2}{r}\right) \leq 0$.

The limiting cases $G_k \in]\theta_1, \theta_2]$, $G_k = \theta_2$ and $G_k \in [\theta_1, \theta_2[$, $G_k = \theta_1$ are treated similarly and (5.5) is proved.

In order to prove the existence and uniqueness of a strong solution, we first prove pathwise uniqueness and then the existence of a weak solution. Let $T > 0$, $t \in [0, T]$, n be fixed and $\phi(s) := \phi_n(s)$, $R_t := R_t^{n,A^\star}$, $\Theta_t := \Theta_t^{n,A^\star}$. Let (R_t, Θ_t) and (R'_t, Θ'_t) be two solutions of the system (5.4). By Itô's Lemma,

$$\begin{aligned}
&\mathbb{E}\left((R_t - R'_t)^2 + (\Theta_t - \Theta'_t)^2\right) \qquad (5.9)\\
&= 2\mathbb{E}\left(\int_0^t [(R_s - R'_s)(\phi(R_s)\mu_1(\Theta_s) - \phi(R'_s)\mu_1(\Theta'_s))\right.\\
&\quad \left. + (\Theta_s - \Theta'_s)(\phi(R_s)\mu_2(R_s, \Theta_s) - \phi(R'_s)\mu_2(R'_s, \Theta'_s))]ds\right)\\
&\quad + 2\mathbb{E}\left(\int_0^t (\Theta_s - \Theta'_s)(\phi(R_s) - \phi(R'_s))\sigma dW_s\right) + \mathbb{E}\left(\int_0^t (\phi(R_s) - \phi(R'_s))^2\sigma^2 ds\right).
\end{aligned}$$

Using the monotonicity property (5.5), the Lipschitz property of ϕ and the fact that the stochastic integral is a martingale, for a positive constant C, (5.9) is bounded from above by

$$C\int_0^t \mathbb{E}\left((R_s - R'_s)^2 + (\Theta_s - \Theta'_s)^2\right) ds. \qquad (5.10)$$

In order to prove the martingale property, since ϕ is bounded, it suffices to show that

$$\mathbb{E}\left(\int_0^T (\Theta_t)^2 dt\right) < \infty. \qquad (5.11)$$

From (5.4) it follows that

$$\begin{aligned}
\mathbb{E}\left(\Theta_t\right)^2 &\leq 3\theta_0^2 + 3\mathbb{E}\left(\int_0^t \phi(R_s)\mu_2(R_s, \Theta_s)ds\right)^2 + 3\mathbb{E}\left(\sigma\int_0^t \phi(R_s)dW_s\right)^2\\
&\leq 3\theta_0^2 + 3t\mathbb{E}\left(\int_0^t (\phi(R_s)\mu_2(R_s, \Theta_s))^2 ds\right) + 3\sigma^2\mathbb{E}\left(\int_0^t (\phi(R_s))^2 ds\right) < \infty,
\end{aligned}$$

where we used the Cauchy-Schwartz inequality and the Itô's isometry. This shows that (5.11) is satisfied. From (5.9) and (5.10), by the Grönwall's inequality, for all $t \in [0,T]$,

$$\mathbb{E}\left((R_t - R'_t)^2 + (\Theta_t - \Theta'_t)^2\right) = 0$$

therefore, since the process is continuous, pathwise uniqueness holds.

We now prove the existence of a weak solution on $[0,T]$, for any $T > 0$. Let $(B_t)_{t\in[0,T]}$ be a standard Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P}_{r,\theta})$. Then the following system has Lipschitz and bounded coefficients:

$$\begin{cases} dR_t = \phi(R_t)\mu_1(\Theta_t)dt, \\ d\Theta_t = \sigma dB_t. \end{cases} \tag{5.12}$$

The existence of a unique strong solution to (5.12), for $t \in [0,T]$, follows from standard results, see for example [6, Chap. 5, Thm 2.9]. Since the function $\phi(r)\mu_2(r,\theta)$ is bounded for $(r,\theta) \in [0,\infty[\times\mathbb{R}$, then Novikov's condition implies that the process

$$Z_t = \exp\left\{-\frac{1}{2}\int_0^t \frac{[\phi(R_t)\mu_2(R_s,\Theta_s)]^2}{\sigma}ds + \int_0^t \frac{\phi(R_t)\mu_2(R_s,\Theta_s)}{\sigma}dB_s\right\}, \qquad t \in [0,T],$$

is a martingale. Let $\mathbb{Q}_T$ be the probability measure on $(\Omega, \mathcal{F}_T)$ defined as $d\mathbb{Q}_T = Z_T d\mathbb{P}_{r,\theta}$. By Girsanov's theorem the process

$$dB'_t = dB_t - \int_0^t \phi(R_t)\mu_2(R_s,\Theta_s)ds, \qquad t \in [0,T],$$

is a $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q}_T)$ Brownian motion. Under this Brownian motion, system (5.12) becomes

$$\begin{cases} dR_t = \phi(R_t)\mu_1(\Theta_t)dt, \\ d\Theta_t = \phi(R_t)\mu_2(R_t,\Theta_t) + \sigma dB'_t, \end{cases}$$

so that (R_t, Θ_t) is a weak solution to (5.4) in $[0,T]$. Weak existence and strong uniqueness imply that there exists a unique strong solution to (5.4) in $[0,T]$.

It remains to prove is that $\tau_\eta^{A^\star} < \sqrt{2}\frac{(R_0-\eta)}{v}$ for every $\eta > 0$. The process R_t satisfies the equation

$$R_{\tau_\eta^{A^\star}} = R_0 + \int_0^{\tau_\eta^{A^\star}} \phi(R_t)\mu_1(\Theta_t)dt.$$

If $R_t \geq \eta$, then $\phi(R_t) = 1$ and, by (5.2) and (5.3), it follows that $\mu_1(\theta) \leq -\frac{v}{\sqrt{2}}$ for every $\theta \in \mathbb{R}$. Therefore

$$\eta = R_{\tau_\eta^{A^\star}} = R_0 + \int_0^{\tau_\eta^{A^\star}} \mu_1(\Theta_t)dt \leq R_0 - \frac{v}{\sqrt{2}}\tau_\eta^{A^\star},$$

so

$$\tau_\eta^{A^\star} \leq \sqrt{2}\frac{(R_0-\eta)}{v}.$$

Since $\tau_\eta^{A^\star}$ is bounded and (5.4) admits a strong solution in $[0,T]$, for any $T > 0$, it follows that it also admits a strong solution in $[0,\tau_\eta^{A^\star}]$ and this completes the proof. □

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